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Dynamic Pricing with Demand Learning and Reference Effects

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Abstract. We consider a seller's dynamic pricing problem with demand learning and reference effects. We first study the case in which customers are loss-averse: they have a *reference price* that can vary over time, and the demand reduction when the selling price exceeds the reference price dominates the demand increase when the selling price falls behind the reference price by the same amount. Thus, the expected demand as a function of price has a time-varying "kink" and is not differentiable everywhere. The seller neither knows the underlying demand function nor observes the time-varying reference prices. In this setting, we design and analyze a policy that (i) changes the selling price very slowly to control the evolution of the reference price and (ii) gradually accumulates sales data to balance the trade-off between learning and earning. We prove that, under a variety of reference-price updating mechanisms, our policy is asymptotically optimal; that is, its T-period revenue loss relative to a clairvoyant who knows the demand function and the reference-price updating mechanism grows at the smallest possible rate in T. We also extend our analysis to the case of a fixed reference price and show how reference effects increase the complexity of dynamic pricing with demand learning in this case. Moreover, we study the case in which customers are gain-seeking and design asymptotically optimal policies for this case. Finally, we design and analyze an asymptotically optimal statistical test for detecting whether customers are loss-averse or gain-seeking.

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1. Introduction

1.1. Background and Motivation

In dynamic pricing applications, the demand for a product is influenced not only by the current selling price, but also by customers' price expectation. As the customers observe the prices charged for the product, they form a *reference price* that represents their price expectation based on the price history. Given this reference point, the customers perceive price increases and decreases relative to the reference price as "losses" and "gains," respectively. Thus, the demand for the product depends on both the current selling price and losses/gains quantified by the difference between the reference price and the current selling price. The literature on the behavioral foundation of reference prices dates back to the seminal work of Kahneman and Tversky on prospect theory (Kahneman and Tversky 1979, Tversky and Kahneman 1991). This theory predicts that customers respond to losses and gains asymmetrically. As empirically validated in various studies, customers are typically *loss-averse* in pricing contexts (see, e.g., Kalyanaram and Winer 1995, Meyer and Johnson 1995, and the references therein)—that is, the loss perceived when the current price exceeds the reference price by some margin is greater than the gain perceived when the reference price exceeds the current price by the same margin. Viewing the relationship between demand and price as a *demand function* that maps the current selling price to the expected demand under that price, a natural consequence of the aforementioned asymmetry is a nonstationary "kink" in the demand function. That is, this function is nondifferentiable when the current selling price equals the reference price, which can vary over time as the customers update their price expectation.

Accounting for reference prices is crucial in pricing practice. In the retail industry, in which ample evidence on reference effects is observed, there is a plethora of firms enjoying multibillion-dollar revenues per year (Chen et al. 2017). Thus, any improvement in a pricing strategy is extremely valuable in terms of

revenues. In this context, the extant evidence suggests that it is essential to incorporate reference-price effects into a demand model. Analyzing household-level consumption data to investigate reference effects, Rajendran and Tellis (1994) report that ignoring reference prices results in suboptimal pricing. Greenleaf (1995) presents empirical and analytical results showing how reference effects impact a firm's profits from price promotions and finds that judicious management of reference prices can substantially increase profits. Furthermore, Mazumdar et al. (2005) review several empirical studies providing strong evidence of reference-price effects for frequently purchased products and suggest that reference prices should be taken into account in managerial decisions on pricing and promotions. In a more recent study, Cornelsen et al. (2018) show evidence on reference-price effects in a household-level data set of food purchases and conclude that firms should account for reference effects in their demand models to avoid serious biases in pricing strategies.

Based on these findings, reference prices and their formation significantly affect a seller's revenue and play an important role in dynamic pricing decisions. Accordingly, reference prices have attracted a lot of interest in the literature on behavioral economics, marketing, and operations management (for an overview of the vast literature on reference prices, see Kalyanaram and Winer 1995, Mazumdar et al. 2005, Arslan and Kachani 2010, and the references therein). In particular, the question of how a seller should optimally price its products in view of this reference-price effect has generated a large stream of research (see Section 1.3 for a review of related literature). In this line of research, it is typically assumed that the demand function is completely known to the seller. This assumption obviously hinders effective application of the resulting pricing policies in practice, where demand functions are typically unknown and have to be learned from sales data. Notably, the emergence of online retail channels significantly facilitates the use of digital sales data in pricing decisions. As in conventional pricing settings, it is essential to take reference-price effects into consideration in online pricing settings, which include online grocery stores (Sheehan et al. 2019), online travel shopping websites (Nieto-García et al. 2017), online hardware retailers (Ray et al. 2015), and online bookstores (Gupta and Kim 2010, Kim et al. 2012). The aforementioned practical settings motivate the current paper, in which we study dynamic pricing in the presence of reference-price effects and incomplete information about the demand function.

1.2. Main Contributions and Managerial Insights

Our study makes four contributions to the dynamic pricing literature. In what follows, we describe these contributions.

1.2.1. How to Implement Dynamic Pricing Under Model Uncertainty and Reference Effects. Our work sheds light on the design and implementation of dynamic pricing and demand learning in the presence of reference effects. Following the antecedent literature (Sorger 1988, Winer 1988, Greenleaf 1995, Kopalle et al. 1996, Fibich et al. 2003, Popescu and Wu 2007), we initially consider a base setting in which customers are loss-averse and reference prices are updated via an exponential smoothing (ES) mechanism. We introduce model uncertainty into this widely used setting by considering a seller who knows neither the underlying demand function nor the parameters of the reference-price updating mechanism. We show that the cost of neglecting reference effects is low if the seller has perfect information about the demand model, but the same cost may be very high in the presence of demand model uncertainty (see Section 3.1). In light of this, we design a dynamic pricing-and-learning policy that judiciously accounts for reference effects. Because price experiments yield unobservable fluctuations in reference prices, our policy carefully stabilizes the reference-price process by changing the selling price very slowly. Measuring performance by *regret*, that is, the T-period revenue loss relative to a clairvoyant who knows all model parameters, we prove that our "slow-moving" pricing policy is asymptotically optimal in the sense that it achieves the smallest growth rate of regret (see Section 3.2).

1.2.2. Extension to More General Reference-Price Updating Mechanisms. The asymptotic optimality of our slow-moving pricing policy is a robust result that does not heavily rely on the exponential smoothing reference-price formation. We show that our performance guarantee for this policy is valid for a fairly general family of reference-price updating mechanisms (see Section 3.3). Antecedent results obtained in the case of full information on the demand model typically depend on the assumption of exponential smoothing reference-price formation, which is difficult to verify in practice. In particular, when the seller has incomplete information about the demand model and cannot directly observe the underlying reference-price process, it is challenging to check the validity of this assumption. Our generalized analysis indicates that the aforementioned slow-moving pricing policy performs well even when reference prices do not necessarily follow an exponential smoothing updating mechanism.

1.2.3. Investigating the Case of a Fixed Reference Price. We analyze a case motivated by settings in which the seller has been charging an incumbent price for a long time. Given the extensive historical use of the incumbent price in this case, the seller has learned the

expected demand at this price, and the customers have settled their reference price at the same price point. Our analysis reveals that reference effects substantially increase the complexity of this incumbent-price formulation: without reference effects, it is possible to achieve a regret of order $\log T$ in this setting (Keskin and Zeevi 2014), but in the presence of reference effects, we prove that the regret of any admissible policy has to be at least in the order of $\sqrt{T}$ (see Section 4.2). To further demonstrate the added complications in learning resulting from reference effects, we examine the performance of passive learning policies in this setting. In the incumbent-price formulation with no reference effects, passive learning policies exhibit incomplete learning with a positive probability less than one (Keskin and Zeevi 2018), whereas with reference effects, they suffer from incomplete learning *with probability one* (see Section 4.4). This difference shows how reference effects can exacerbate the incomplete learning complication of passive learning policies. Our incomplete learning result implies that the regret of a passive learning policy is linear in T, thereby emphasizing the need for price experimentation in this context. Accordingly, we construct a dynamic pricing-and-learning policy that achieves asymptotic optimality with a regret of order $\sqrt{T}$ in this case (see Section 4.3). To provide a unifying approach, we also design a statistical test that identifies whether the underlying reference price is fixed or not without increasing the growth rate of regret (see Online Appendix E).

1.2.4. Extension to Gain-Seeking Customer Behavior. We generalize our analysis to a case in which customers are gain-seeking (instead of being loss-averse), and our analysis of this case brings forth a new set of results. In this case, if the seller has perfect information about the demand model, it is asymptotically optimal to use a cyclic price-skimming strategy. Unlike the case of loss-averse customers, the smallest growth rate of regret for gain-seeking customers depends on unknown model parameters—specifically, it is based on the optimal cycle length, which is parameter-dependent. Building on this observation, we design a dynamic pricing-and-learning policy that achieves asymptotic optimality in this case (see Section 5). We also unify our analyses for the cases of loss-averse and gain-seeking customers by designing a statistical test that identifies whether customers are loss-averse or not without increasing the growth rate of regret (see Section 6).

The aforementioned contributions lead to key managerial insights for the practice of dynamic pricing with demand learning. First, when customers are loss-averse and reference prices are varying over time, it is crucial for a seller to slowly vary the selling price to avoid excessive fluctuations in the reference-price process. If the seller does not carefully control how prices vary over time, the revenue performance may severely deteriorate. Moreover, the idea of moving prices slowly works well under a wide variety of reference-price updating mechanisms. Second, even when the customers' reference price is fixed at a historically incumbent price, the seller's dynamic pricing problem is as challenging as in the case of time-varying reference prices, and the seller's revenue performance in these two cases is similar. Third, when customers are gain-seeking, the seller needs to adopt a different approach by using a cyclic price-skimming strategy, and in this case, the seller's revenue performance depends on the optimal cycle length. Fourth, we also provide a unifying approach to dynamic pricing with demand learning and reference effects. Using our statistical tests to detect whether customers are loss-averse or gain-seeking and whether the reference price is fixed or varying over time, the seller can efficiently identify the ambient reference-price setting, which could be any of the settings studied in Sections 3–5. After that, the seller should use the pricing policy we design for that setting. We hope that these qualitative insights provide useful guidelines for designing dynamic pricing-and-learning policies in practice.

To derive our results, we develop new proof techniques that are distinct from those in the existing studies on dynamic pricing with demand learning. First, as our discrete-time learning-and-earning problem defies exact solution, we rigorously connect our problem to a continuous-time problem that has a closed-form solution. Our analysis explicitly characterizes the potential losses resulting from the continuous-time approximation as well as the discrete-time reference-price dynamics (see the proofs in Online Appendix A). Second, when customers are gain-seeking, reference prices necessitate the use of cyclical price skimming. We analyze the optimal cyclical policies in this case and characterize their performance (see the proofs in Online Appendix C). As part of this analysis, we investigate the asymptotic optimality of cyclical policies for gain-seeking customers; these results could be of independent interest in the study of dynamic pricing with reference effects. Third, we design and analyze statistical tests that efficiently identify whether customers are loss-averse or gain-seeking and whether the reference price is fixed or time-varying (see the proofs in Online Appendices D and E). These analysis features differentiate our proofs from those in the antecedent literature on dynamic pricing and learning.

1.3. Related Literature

This paper contributes to two streams of literature: dynamic pricing with reference-price effects and dynamic pricing with incomplete demand information.

Optimal dynamic pricing with reference-price effects has been studied extensively in recent years. The literature up to 2010 is summarized by Arslan and Kachani (2010), whereas Nasiry and Popescu (2011), Heidhues and Koszegi (2014), Wu et al. (2015), Chevanaz (2016), Wang (2016), Chen et al. (2017), Bi et al. (2017), and Yang et al. (2017) constitute a sample of more recent papers that study properties of optimal pricing policies under various modeling assumptions about the reference-price process. A common assumption in this stream of literature is that the seller knows the demand function. In this paper, we depart from this assumption and consider dynamic pricing with reference-price effects in which the seller has to *learn* the demand function from accumulating sales data.

There is a large and expanding literature on such learning-and-earning problems; see, for example, Araman and Caldentey (2009), Besbes and Zeevi (2009), Farias and van Roy (2010), Harrison et al. (2012), Broder and Rusmevichientong (2012), den Boer and Zwart (2014), Keskin and Zeevi (2014), Keskin (2014), and den Boer and Zwart (2015). For a recent survey, we refer to den Boer (2015a). With the exception of the recent working paper by Kazerouni and van Roy (2017), the effect of reference prices has not been considered in this literature. Kazerouni and van Roy (2017) consider a pricing problem in which the sales horizon is divided into "episodes" of equal finite length. The expected demand is a function of the n most recent selling prices within the current episode, in which n is a finite memory constant known to the seller. In this setting, the authors propose and analyze a pricing algorithm based on Thompson sampling. An important distinction between this work and ours is the finite-memory assumption. Under this assumption, prices older than n periods or prices from past episodes (no matter how recent they are) do not influence the demand; that is, customers' memory is limited and periodically eliminated. This excludes, for example, the exponential smoothing mechanism that is typically employed for reference-price updating in the related literature. In contrast, our work presents an asymptotically optimal pricing policy that is valid for a large class of reference-price updating mechanisms, including exponential smoothing. Another difference is that Thompson sampling typically requires one to assume a conjugate pair of demand and belief distributions; this is not necessary in our approach.

Our work is also related to dynamic pricing with covariates (see, e.g., Qiang and Bayati 2016, Javanmard and Nazerzadeh 2019, Cohen et al. 2020, Ban and Keskin 2021). In this literature, the expected demand in a single time period t is a function of both the current selling price and some feature vector x_t. The feature vectors $\{x_t\}$ either follow a sequence of independent and identically distributed (i.i.d.) random variables (as in Qiang and Bayati 2016, Javanmard and Nazerzadeh 2019, Ban and Keskin 2021) or are adversarially selected (as in Cohen et al. 2020). The reference-price sequence in our model can also be regarded as a covariate sequence; however, it is neither an i.i.d. sequence nor adversarially selected by nature, but *endogenously* determined by the price history. This distinguishes our work from this stream of literature.

There is also a connection between our work and the literature on dynamic pricing and demand learning with time-varying demand functions (see, e.g., den Boer 2015b, Keskin and Zeevi 2016). In this literature, the changes in the demand function are exogenously driven, whereas in our setting, the changes in the demand function are endogenously influenced by seller's pricing decisions.

1.4. Organization of the Paper

The rest of this paper is organized as follows. In Section 2, we provide our problem formulation and notation. In Section 3, we analyze our base setting, in which customers are loss-averse and reference prices are varying over time. In Section 4, we study the case of a fixed reference price. In Section 5, we extend our analysis from the case of loss-averse customers to the case of gain-seeking customers. In Section 6, we provide a statistical test that identifies whether customers are loss-averse or gain-seeking without deteriorating asymptotic performance. Our concluding remarks are provided in Section 7, and all mathematical proofs are presented in the appendices.

2. Problem Formulation

Consider a firm, referred to as the *seller*, that sells a product over $T \in \mathbb{N}$ discrete time periods indexed by $t = 1, 2, \ldots, T$. At the beginning of each period t, the seller sets a price $p_t \in [\underline{p}, \overline{p}]$ for its product, where $\underline{p}$ and $\overline{p}$ are the smallest and largest feasible prices satisfying $0 \le \underline{p} < \overline{p} < \infty$. After setting the price, the seller observes the demand d_t for the product and collects revenue $p_t d_t$. The demand is influenced by (i) the selling price, (ii) customers' reference price, and (iii) temporal demand shocks. Given the selling price p_t, the demand in period t is

$$d_t = d(p_t, r_t) + \epsilon_t \quad \text{for} \quad t = 1, \ldots, T, \tag{2.1}$$

where $r_t \in [\underline{p}, \overline{p}]$ is the reference price at the beginning of period t; $\{\epsilon_t\}_{t \in \mathbb{N}}$ are unobservable demand shocks, which follow a sequence of independent and identically distributed random variables with a distribution unknown to the seller and with zero mean, finite variance, and a well-defined moment-generating function (i.e., there exists $m_0 > 0$, not necessarily known by the seller, such that $\mathbb{E}[\exp(m\epsilon_1)] < \infty$ for

all m with $|m| \le m_0$); and $d(\cdot,\cdot)$ is the demand function given by

$$d : [\underline{p}, \overline{p}]^2 \to \mathbb{R}_+,$$

$$d(p,r) = \alpha - \beta p + \eta^+(r-p)^+ - \eta^-(p-r)^+ = \begin{cases} \alpha - \beta p + \eta^+(r-p) & \text{if } p < r, \\ \alpha - \beta p - \eta^-(p-r) & \text{if } p \ge r, \end{cases} \tag{2.2}$$

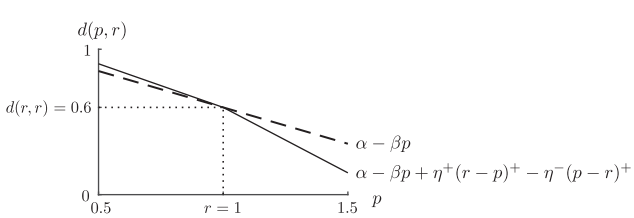
with $\alpha \ge 0$, $\beta > 0$, $\eta^+ \ge 0$, and $\eta^- \ge 0$. In (2.2), α, β, η^+, and η^- are demand parameters that are initially unknown to the seller. The parameters α and β represent the market size and the customers' price sensitivity, respectively. On the other hand, η^+ and η^- represent the reference effects: if the selling price p is less than the reference price r, then the expected demand increases by $\eta^+(r-p)$, but if p exceeds r, then the expected demand decreases by $\eta^-(p-r)$. Figure 1 displays a demand function example.

Based on the strong empirical evidence on loss aversion in pricing contexts (see, e.g., Kalyanaram and Winer 1995, Meyer and Johnson 1995), we first consider the case in which customers are loss-averse: the positive impact of charging a price lower than the reference price is less than the negative impact of exceeding the reference price; that is, $\eta^+ \le \eta^-$. (For the sake of completeness, we later relax the loss-aversion assumption and study the case of gain-seeking customers in Section 5. We also provide a statistical test that can efficiently identify whether customers are loss-averse or gain-seeking; see Section 6.) Note that the demand function in (2.2) is an aggregate demand model, and we describe loss aversion and gain seeking directly on this aggregate model. Thus, throughout the paper, we use the term "loss-averse (gain-seeking) customers" to refer to the case of $\eta^+ \le \eta^-$ ($\eta^+ > \eta^-$).

We denote by $\theta := (\alpha, \beta, \eta^+, \eta^-)$ the vector of unknown demand parameters and assume that θ lies in a compact set Θ defined as follows:

$$\Theta := \{(\alpha,\beta,\eta^+,\eta^-) \in [\alpha_{\min},\alpha_{\max}] \times [\beta_{\min},\beta_{\max}] \times [\eta^+_{\min},\eta^+_{\max}] \times [\eta^-_{\min},\eta^-_{\max}] : \eta^+ \le \eta^-\}, \tag{2.3}$$

Figure 1. Demand Function Example



Notes. The bold curve displays an example of the demand function (2.2) with parameters $\alpha = 1.1$, $\beta = 0.5$, $\eta^+ = 0.1$, and $\eta^- = 0.4$. The dashed curve shows the mapping $p \longmapsto \alpha - \beta p$, which corresponds to the demand function in the absence of reference effects.

where $\alpha_{\min}$, $\alpha_{\max}$, $\beta_{\min}$, $\beta_{\max}$, $\eta^+_{\min}$, $\eta^+_{\max}$, $\eta^-_{\min}$, $\eta^-_{\max}$, are known constants that satisfy $[\alpha_{\min},\alpha_{\max}] \times [\beta_{\min},\beta_{\max}] \times [\eta^+_{\min},\eta^+_{\max}] \times [\eta^-_{\min},\eta^-_{\max}] \subset [0,\infty) \times (0,\infty) \times [0,\infty) \times [0,\infty)$. As explained earlier, the condition that $\eta^+ \le \eta^-$ represents the loss aversion of customers in the definition of Θ. To emphasize the dependence of the demand function on θ, we occasionally use the notation $d(p,r,\theta)$ instead of $d(p,r)$. We assume that, for all $\theta \in \Theta$, the expected demand $d(p,r,\theta)$ is nonnegative for all $(p,r) \in [\underline{p}, \overline{p}]^2$; that is, $\alpha - \beta\overline{p} - \eta^-(\overline{p} - \underline{p}) \ge 0$ for all $\theta = (\alpha,\beta,\eta^+,\eta^-) \in \Theta$. Furthermore, we also assume that the optimal price in the absence of reference effects, namely, the maximizer of the mapping $p \longmapsto p(\alpha - \beta p)$, lies in the interior of the feasible price set; that is, $\alpha/(2\beta) \in (\underline{p}, \overline{p})$ for all $\theta = (\alpha,\beta,\eta^+,\eta^-) \in \Theta$.

The dynamics of the reference-price process $\{r_t\}_{t=1,\ldots,T}$ are as follows:

$$r_t = \rho_t(p_1,\ldots,p_{t-1}) \quad \text{for } t = 2,\ldots,T, \tag{2.4a}$$

$$r_1 = \rho_1, \tag{2.4b}$$

where $\rho_1 \in [\underline{p}, \overline{p}]$ denotes the initial reference price at the start of the sales horizon and ρ_t is a continuous function from $[\underline{p},\overline{p}]^{t-1}$ to $[\underline{p},\overline{p}]$ for $t = 2,\ldots,T$. We call $\rho = \{\rho_t\}_{t\in\mathbb{N}}$ the *reference-price updating mechanism*. The seller does not necessarily know the exact reference-price updating mechanism ρ but knows that it is an element of a nonempty set $\mathcal{R}$ such that

$$\mathcal{R} \subset \{\{f_t\}_{t\in\mathbb{N}} : f_1 \in [\underline{p},\overline{p}] \text{ and } f_t \in \mathcal{C}([\underline{p}, \overline{p}]^{t-1}, [\underline{p}, \overline{p}]) \text{ for all } t \ge 2\},$$

where $\mathcal{C}([\underline{p}, \overline{p}]^{t-1}, [\underline{p}, \overline{p}])$ denotes the set of continuous functions from $[\underline{p}, \overline{p}]^{t-1}$ to $[\underline{p}, \overline{p}]$. The set $\mathcal{R}$ allows the seller to incorporate structural assumptions about the evolution of reference prices. For example, if the seller believes that the reference-price process is generated via exponential smoothing, then the seller could assume that there exists a (possibly unknown) constant $\zeta \in [0,1]$ such that, for all $\{f_t\}_{t\in\mathbb{N}} \in \mathcal{R}$,

$$f_{t+1}(p_1,\ldots,p_t) = \zeta f_t(p_1,\ldots,p_{t-1}) + (1-\zeta)p_t \quad \text{for} \quad t = 1,2,\ldots,$$

where we write $f_1(\emptyset) = f_1$. In subsequent sections, we consider several different instances for $\mathcal{R}$.

An admissible pricing policy π is a collection of mutually independent random variables,

$$\pi = \{\pi(H) : t = 1,\ldots,T,\ H \in ([\underline{p},\overline{p}] \times \mathbb{R}_+)^{t-1}\}, \tag{2.5}$$

defined on a common probability space such that $\pi(H) \in [\underline{p},\overline{p}]$ almost surely for all H and π contains an element $\pi(\emptyset)$. We say that the seller uses the pricing policy π or, equivalently, that prices are generated under π if

$$p_1 = \pi(\emptyset) \text{ almost surely,} \tag{2.6a}$$

$$p_t = \pi(p_1, d_1, \ldots, p_{t-1}, d_{t-1}) \text{ almost surely for } t = 2, \ldots, T. \tag{2.6b}$$

We emphasize that, according to the definition of admissible pricing policies, the reference price is not observable to the seller.

The performance of a pricing policy π is measured by its regret: the expected difference between the cumulative revenue obtained by a seller who uses π and the cumulative revenue obtained by a revenue-maximizing clairvoyant who knows with certainty the underlying reference-price updating mechanism ρ and demand parameter vector θ. We denote the expected single-period revenue function by $R(p, r, \theta) := p\,d(p, r, \theta)$ for all $(p, r) \in [\underline{p}, \overline{p}]^2$ and $\theta \in \Theta$. The optimal expected cumulative revenue under full information as a function of $T \in \mathbb{N}$, $\rho \in \mathcal{R}$, and $\theta \in \Theta$ is given by

$$\Pi^*(T, \rho, \theta) := \sup_{(p_1, \ldots, p_T) \in [\underline{p}, \overline{p}]^T} \left\{ \sum_{t=1}^{T} R(p_t, \rho_t(p_1, \ldots, p_{t-1}), \theta) \right\}; \tag{2.7}$$

here and in the rest of this paper, we write $\rho_1(\emptyset) := \rho_1$ for the initial reference price. The regret of π is then defined as

$$\Delta_\pi(T, \rho, \theta) = \Pi^*(T, \rho, \theta) - \mathbb{E}_\pi\left[\sum_{t=1}^{T} R(p_t, r_t, \theta)\right], \tag{2.8}$$

where $p_1, \ldots, p_T$ satisfy (2.6) and $r_1, \ldots, r_T$ satisfy (2.4), and $\mathbb{E}_\pi[\cdot]$ denotes the expected value under policy π. The subscript π in the expectation operator $\mathbb{E}_\pi[\cdot]$ emphasizes the dependence of $p_1, \ldots, p_T$ on π. Accordingly, we write $\mathbb{P}_\pi\{\cdot\}$ for probabilities that depend on π (via $p_1, \ldots, p_T$).

Remark 1. Our performance measure, regret, is perhaps the most commonly used performance measure in the literature on dynamic pricing with demand learning (see, e.g., Broder and Rusmevichientong 2012, Harrison et al. 2012, den Boer and Zwart 2014, Keskin and Zeevi 2014). As detailed, the T-period regret of a policy measures that policy's cumulative expected revenue loss after T time periods. It is also possible to measure a policy's performance by its cumulative discounted loss over an infinite time horizon (see, e.g., Popescu and Wu 2007). To compare our results with the antecedent literature on dynamic pricing and learning, we adopt the standard performance measure in that literature, namely, regret.

3. Learning and Earning Under Time-Varying Reference Prices

In this section, we design and analyze a pricing policy for the case of time-varying reference prices. Following the literature on reference-price effects (see, e.g., Sorger 1988, Winer 1988, Greenleaf 1995, Kopalle et al. 1996, Fibich et al. 2003, Popescu and Wu 2007), we initially consider reference prices that are formed via ES; that is,

$$\rho \in \mathcal{R}_{\text{ES}} := \cup_{\zeta \in [0,1)} \mathcal{R}_\zeta, \tag{3.1}$$

where

$$\mathcal{R}_\zeta = \left\{ \begin{array}{l} \{\rho_t\}_{t \in \mathbb{N}} : \rho_{t+1}(p_1, \ldots, p_t) = \zeta \rho_t(p_1, \ldots, p_{t-1}) + (1-\zeta) p_t \\ \text{for all } t \in \mathbb{N}, \{p_t\}_{t \in \mathbb{N}} \in [\underline{p}, \overline{p}]^{\mathbb{N}}, \text{ and } \rho_1 \in [\underline{p}, \overline{p}] \end{array} \right\}$$

$$\text{for all } \zeta \in [0, 1). \tag{3.2}$$

In this formulation, the parameter ζ, usually referred to as the exponential smoothing coefficient, captures the rate at which the reference price evolves: if $\zeta = 0$, then the reference price moves very quickly in response to price changes; it is always equal to the price in the preceding period. But, if ζ is close to one, then the reference price moves more slowly. We emphasize that according to the formulation in (3.1) and (3.2), the seller does not know the value of $\zeta \in [0, 1)$. In Section 3.3, we extend the formulation in (3.1) and (3.2) to more general time-varying reference-price updating mechanisms. In Section 4, we study a case in which $\zeta = 1$ and, thus, the reference does not change over time.

3.1. Can Reference-Price Effects Be Safely Ignored in Dynamic Learning?

A natural question on reference-price effects is whether the seller can safely ignore them in the context of dynamic pricing with demand learning. This section focuses on this question.

If reference effects are neglected, the single-period expected revenue equals $p(\alpha - \beta p)$, and hence, the optimal selling price is $\alpha/(2\beta)$. Interestingly, when reference effects are present and updated via exponential smoothing, it turns out that fixing the selling price at $\alpha/(2\beta)$ throughout the entire time horizon yields a revenue very close to the revenue under the optimal price sequence: the difference between these two revenues is bounded by a constant independent of the time horizon. To formalize this insight, we first define the aforementioned revenue difference. Given $T \in \mathbb{N}$, $\rho \in \mathcal{R}_{\text{ES}}$, and $\theta \in \Theta$, let $\Pi_{\text{fixed}}(p; T, \rho, \theta)$ be the expected cumulative revenue obtained when $p_1 = \cdots = p_T = p \in [\underline{p}, \overline{p}]$, that is,

$$\Pi_{\text{fixed}}(p; T, \rho, \theta) = \sum_{t=1}^{T} R(p, \rho_t(\underbrace{p, \ldots, p}_{t-1 \text{ times}}), \theta).$$

Moreover, let $\Delta_{\text{fixed}}(p; T, \rho, \theta)$ be the cumulative loss resulting from charging a fixed price $p \in [\underline{p}, \overline{p}]$ as opposed to the best dynamic sequence of prices under full information:

$$\Delta_{\text{fixed}}(p; T, \rho, \theta) = \Pi^*(T, \rho, \theta) - \Pi_{\text{fixed}}(p; T, \rho, \theta).$$

The following result characterizes the cumulative loss $\Delta_{\text{fixed}}(p;T,\rho,\theta)$ for charging the fixed price $p=\alpha/(2\beta)$ in all periods.

Theorem 1 (Loss Under the Fixed Price $\alpha/(2\beta)$). *Let $\rho\in\mathcal{R}_{ES}$ and $\theta=(\alpha,\beta,\eta^+,\eta^-)\in\Theta$. Then, there exists a finite and positive constant c_1* such that

$$\Delta_{\text{fixed}}\left(\frac{\alpha}{2\beta};T,\rho,\theta\right)\le c_1 \textit{ for all } T\in\mathbb{N}. \tag{3.3}$$

We prove Theorem 1 by establishing a connection between our discrete-time total-revenue setting and the continuous-time discounted-revenue setting of Fibich et al. (2003), who give an explicit solution for the optimal price sequence. The constant c_1 depends on $\underline{p}$, $\overline{p}$, Θ, and ζ, and is in the order of $1/(1-\zeta)$. The details of the proof are in Online Appendix A.

Theorem 1 implies that the effect of reference prices can be safely ignored *under full information*, especially if T grows large. It is tempting to think that the same conclusion holds *under demand model uncertainty*, that is, that any pricing-and-learning policy that is asymptotically optimal in the absence of reference effects must also be asymptotically optimal in the presence of reference effects. This, however, is not true. On the contrary, a pricing-and-learning policy that is asymptotically optimal in the absence of reference effects may perform extremely poorly when reference effects are present.

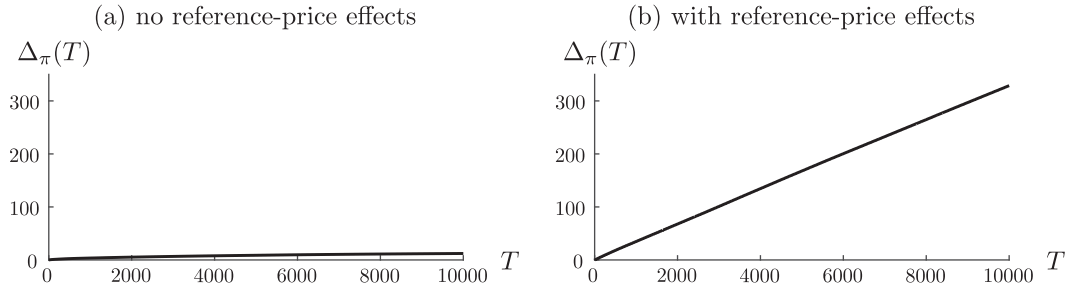
To illustrate this, let us consider a policy called "iterated least squares with deterministic testing" (Keskin and Zeevi 2014), also known as "MLE-CYCLE-S" (Broder and Rusmevichientong 2012). This policy charges a test price $\tilde{p}_1$ in all periods $t\in\{1+2c+\frac{1}{2}c(c+1):c\in\mathbb{N}\cup\{0\}\}$ and a different test price $\tilde{p}_2$ in all periods $t\in\{2+2c+\frac{1}{2}c(c+1):c\in\mathbb{N}\cup\{0\}\}$. In each of the remaining periods, the parameters α and β of a linear demand model $\mathbb{E}[d_t|p_t]=\alpha-\beta p_t$ are estimated based on all available sales data, and based on this, an estimate of the optimal fixed price, namely, $\hat{\alpha}_t/(2\hat{\beta}_t)$, is charged, in which $\hat{\alpha}_t$ and $\hat{\beta}_t$ denote the parameter estimates in period t.

In the absence of reference effects, this policy is asymptotically optimal: its regret after T periods grows in proportion to $\sqrt{T}$, and no policy can have regret growing slower than $\sqrt{T}$ (Keskin and Zeevi 2014). However, reference-price effects eliminate the asymptotic optimality of this policy. Figure 2(a) shows the regret of iterated least squares with deterministic testing in an example without reference effects, and Figure 2(b) shows the regret of this policy in the same example but with reference effects. Whereas the regret of said policy is of order $\sqrt{T}$ in Figure 2(a), it grows linearly in T in Figure 2(b). This illustrates that naively using a policy that is not designed to handle reference-price effects can have a very poor performance.

The reason behind the poor performance of iterated least squares with deterministic testing in the presence of reference effects lies in the estimation of the parameters α and β. Over the periods that are not dedicated to price tests, the underlying reference price moves closer to the estimate of the optimal fixed price, $\hat{\alpha}_t/(2\hat{\beta}_t)$. If parameter estimates are consistent, this term should converge to $\alpha/(2\beta)$. But, as t grows large, a reference-price effect of approximate magnitude $\eta^+(\alpha/(2\beta)-\tilde{p}_i)^+-\eta^-(\tilde{p}_i-\alpha/(2\beta))^+$ is ignored in the periods dedicated to price tests. This results in a misspecified demand model and, thus, inconsistent estimators $\hat{\alpha}_t$ and $\hat{\beta}_t$. Figure 3 illustrates how the model misspecification caused by ignoring reference-price effects leads to inconsistency. As a consequence, the T-period regret of this policy grows linearly in T.

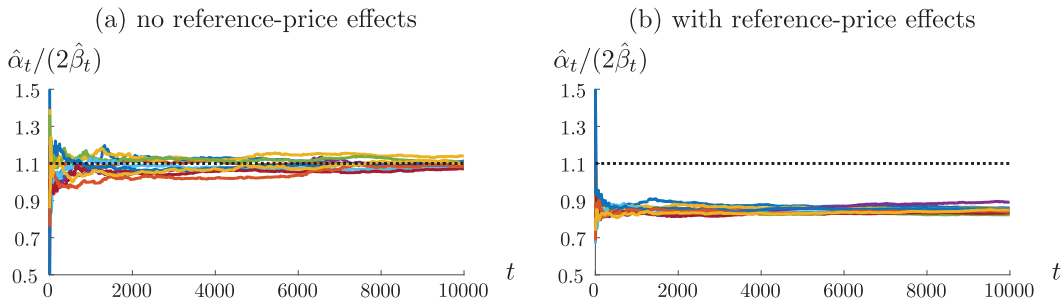
In light of Theorem 1 and Figures 2 and 3, we deduce that, although the cost of ignoring reference-price effects is small when the seller has full information about the demand model, the same cost can be extremely high when the seller faces demand model uncertainty.

Figure 2. Regret Caused by Ignoring Reference-Price Effects Under Demand Model Uncertainty



Notes. (a) The T-period regret of *iterated least squares with deterministic testing* in a setting with no reference-price effects (i.e., $\eta^+=\eta^-=0$). (b) The T-period regret of said policy in the same setting but with reference-price effects. For the reference-price effects in the latter setting, we have $\eta^+=0.1$, $\eta^-=0.3$, and $\zeta=0.1$. In both settings, $\alpha=1.1$, $\beta=0.5$, and the demand shocks ϵ_t are normally distributed with mean zero and standard deviation $\sigma=0.1$. The feasible price interval is $[\underline{p},\overline{p}]=[0.5,1.5]$, and the test prices are $\tilde{p}_1=0.8333$ and $\tilde{p}_2=1.1667$.

Figure 3. (Color online) Pricing While Ignoring Reference-Price Effects Under Demand Model Uncertainty



Notes. The graphs show sample paths of the sequence $\{\hat{\alpha}_t/(2\hat{\beta}_t)\}$, which estimates the optimal fixed price $\alpha/(2\beta)$ under *iterated least squares with deterministic testing* in the settings of Figure 2. (a) is generated in the setting with no reference effects and (b) in the setting with reference effects. The problem parameters are the same as in Figure 2. The optimal fixed price $\alpha/(2\beta)$ equals 1.1 in both settings and is shown in the dotted line in each panel.

3.2. Designing an Asymptotically Optimal Pricing Policy

As established in the preceding section, reference-price effects cannot be safely ignored in the case of demand model uncertainty. The seller needs to account for them while learning the underlying demand model. However, the seller's learning efforts might also complicate reference effects: conducting price tests can cause undesirable fluctuations in the reference-price process. Because reference prices are not observable, large reference-price fluctuations pose a key challenge in the estimation of demand parameters.

To avoid such undesirable fluctuations in reference prices, we design a policy that judiciously stabilizes the reference-price process. To that end, we first note that, if the seller keeps the price fixed for a sufficiently long duration, then the reference price should start to approach this fixed price. Then, the reference effects in (2.2) become relatively small, and the model $\mathbb{E}[d_t|p_t] = \alpha - \beta p_t$ becomes a good approximation to the underlying demand function. To make this intuition work, we need to characterize the convergence rate of reference prices. The following proposition shows that reference prices under an exponential smoothing updating mechanism indeed converge rapidly to a selling price that is kept fixed.

Proposition 1 (Convergence Rate of the Reference Price). *Let $\rho \in \mathcal{R}_{ES}$. Then, there exists a finite and positive constant c_2* such that

$$\sum_{k=s+1}^{t} \left| \rho_k\left(p_1,\ldots,p_s, \underbrace{p,\ldots,p}_{k-s-1 \text{ times}}\right) - p \right| \le c_2 \tag{3.4}$$

for all $s,t \in \mathbb{N}$ satisfying $s < t$, and all $p_1,\ldots,p_s,p \in [\underline{p},\overline{p}]$.

The proof of Proposition 1 is based on the definition of the exponential smoothing updating mechanism. The constant c_2 depends on $\underline{p}$, $\overline{p}$, and ζ, and is in the order of $1/(1-\zeta)$. The details of the proof are in Online Appendix A.

Using the insights of Theorem 1 and Proposition 1, we now construct a policy that achieves the smallest possible growth rate of regret when reference effects are present.

Slow-Moving Pricing Policy $\mathcal{S}(\boldsymbol{n},\boldsymbol{\delta},\hat{a}_0,\hat{b}_0)$
Initialization. Let $\boldsymbol{n} = \{n_i \in \mathbb{N}, i = 1,2,\ldots\}$, $\boldsymbol{\delta} = \{\delta_i \in (0, \frac{1}{2}(\overline{p}-\underline{p})), i = 1,2,\ldots\}$, and $(\hat{a}_0,\hat{b}_0) \in [\alpha_{\min},\alpha_{\max}] \times [\beta_{\min}, \beta_{\max}]$. Set $t_0 = 0$ and $t_i = \sum_{j=1}^{i} 2n_j$ for $i = 1,2,\ldots$ Define

$$\mathcal{T}_i^- = \{t_{i-1}+1,\ldots,t_{i-1}+n_i\} \quad \text{for } i = 1,2,\ldots,$$

$$\mathcal{T}_i^+ = \{t_{i-1}+n_i+1,\ldots,t_{i-1}+2n_i\} \quad \text{for } i = 1,2,\ldots$$

Fix $\hat{p}_0^* = \hat{a}_0/(2\hat{b}_0)$, and for all $i = 1,2,\ldots$ satisfying $t_{i-1} < T$, do the following:
Price Experimentation. Choose

$$p_t = (\hat{p}_{i-1}^* - \delta_i) \vee \underline{p} \quad \text{for all } t \in \mathcal{T}_i^-,$$

$$p_t = (\hat{p}_{i-1}^* + \delta_i) \wedge \overline{p} \quad \text{for all } t \in \mathcal{T}_i^+,$$

where $\vee$ and $\wedge$ denote the maximum and minimum of two numbers, respectively.
Demand Estimation. Compute

$$(\hat{\alpha}_i,\hat{\beta}_i) \in \arg\min_{(\alpha,\beta)} \left\{ \sum_{s \in \mathcal{T}_i^- \cup \mathcal{T}_i^+} (\alpha - \beta p_s - d_s)^2 \right\},$$

and let $\hat{a}_i := \mathcal{P}_{[\alpha_{\min},\alpha_{\max}]}\{\hat{\alpha}_i\}$ and $\hat{b}_i := \mathcal{P}_{[\beta_{\min},\beta_{\max}]}\{\hat{\beta}_i\}$, where $\mathcal{P}_{[\ell,u]}\{x\} := (x \vee \ell) \wedge u$ for $\ell,u,x \in \mathbb{R}$.
Price Optimization. Set

$$\hat{p}_i^* := \frac{\hat{a}_i}{2\hat{b}_i}.$$

Roughly speaking, our policy, called the *slow-moving pricing policy* with parameters $(\boldsymbol{n},\boldsymbol{\delta},\hat{a}_0,\hat{b}_0)$ and denoted by $\mathcal{S}(\boldsymbol{n},\boldsymbol{\delta},\hat{a}_0,\hat{b}_0)$, divides the time horizon into consecutive episodes indexed by $i \in \mathbb{N}$. Throughout the time horizon, $\mathcal{S}(\boldsymbol{n},\boldsymbol{\delta},\hat{a}_0,\hat{b}_0)$ keeps track of an estimate $\hat{p}_{i-1}^*$ of the price $\alpha/(2\beta)$ based on the sales data obtained during the $(i-1)^{\text{st}}$ episode. Here, $\hat{p}_0^* = \hat{a}_0/(2\hat{b}_0)$ is an

initial estimate of $\alpha/(2\beta)$, which could depend on historical information as well as subjective predictions. In the first half of the ith episode, the selling price is set to $(\hat{p}^*_{i-1} - \delta_i) \vee \underline{p}$, and in the second half of the ith episode, the price is set to $(\hat{p}^*_{i-1} + \delta_i) \wedge \overline{p}$, where $\delta_i > 0$. The subtraction and addition of δ_i ensures a minimal amount of price dispersion in each episode, which gradually improves the accuracy in the estimation of $\alpha/(2\beta)$. At the end of the ith episode, the parameters α and β of a linear demand model $\mathbb{E}[d_t|p_t] = \alpha - \beta p_t$ are estimated based on the sales data collected in the ith episode, and given these parameter estimates, a new estimate $\hat{p}^*_i$ of $\alpha/(2\beta)$ is computed. As the notation $\mathcal{S}(\boldsymbol{n}, \boldsymbol{\delta}, \hat{a}_0, \hat{b}_0)$ implies, the episode lengths $\boldsymbol{n} = \{n_i, i = 1, 2, \ldots\}$, the price perturbations $\boldsymbol{\delta} = \{\delta_i, i = 1, 2, \ldots\}$, and the initial estimates $(\hat{a}_0, \hat{b}_0)$ are input parameters of the policy.

The next result provides a performance guarantee for the slow-moving pricing policy.

Theorem 2 (Asymptotic Optimality of the Slow-Moving Pricing Policy). *Let $\rho \in \mathcal{R}_{ES}$, $\theta \in \Theta$, and $\pi = \mathcal{S}(\boldsymbol{n}, \boldsymbol{\delta}, \hat{a}_0, \hat{b}_0)$, where $n_i = 2^i$ and $\delta_i = c_0 2^{-i/4}$ for all $i \in \mathbb{N}$, $c_0 \in (0, 2^{-3/4}(\overline{p} - \underline{p}))$, and $(\hat{a}_0, \hat{b}_0) \in [\alpha_{\min}, \alpha_{\max}] \times [\beta_{\min}, \beta_{\max}]$. Then, there exists a finite and positive constant c_3 such that*

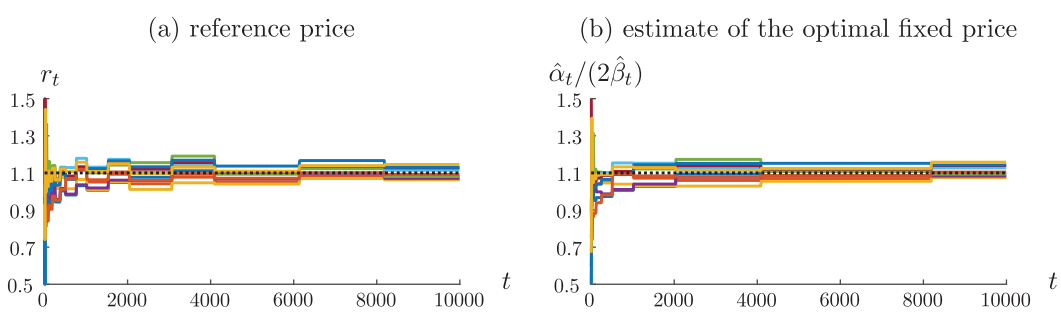
$$\Delta_\pi(T, \rho, \theta) \le c_3 \sqrt{T} \quad \textit{for all } T \in \mathbb{N}.$$

Theorem 2 states that the T-period regret of the slow-moving pricing policy grows at most in the order of $\sqrt{T}$. As mentioned, this is the smallest achievable growth rate of regret even in the case of no reference-price effects (see, e.g., Keskin and Zeevi 2014, theorem 1). Because the absence of reference effects is a special case of our problem setting in which $\eta^+ = \eta^- = 0$, Theorem 2 implies that the slow-moving pricing policy is *asymptotically optimal* in the sense that its regret increases at the best possible rate.

In the proof of Theorem 2, we establish that the slow-moving pricing policy $\mathcal{S}(\boldsymbol{n}, \boldsymbol{\delta}, \hat{a}_0, \hat{b}_0)$ avoids excessive fluctuations in the reference-price process by changing the price only after an exponentially growing number of periods $n_i = 2^i$—hence, the name "slow-moving" pricing policy. Figure 4 illustrates how the reference price and the estimate of the optimal fixed price evolve very gradually under this policy. Intuitively, slowly moving the price enables the reference-price process to converge rapidly and exhibit only a small of variation in each episode during which the price is kept fixed, as characterized by Proposition 1; see Figure 4(a). Consequently, $p_t - r_t$ converges to zero as t increases, the effect of reference prices on the demand diminishes, letting the parameter estimate vector $(\hat{\alpha}_i, \hat{\beta}_i)$ converge to (α, β), and the price estimate $\hat{p}^*_i = \hat{\alpha}_i/(2\hat{\beta}_i)$ to $\alpha/(2\beta)$ as i grows large; see Figure 4(b). By Theorem 1, the loss caused by using $\alpha/(2\beta)$ throughout the time horizon instead of the best dynamic sequence of prices is bounded, and by Proposition 1, if the selling price is fixed, then the cumulative absolute deviation of the reference price from the fixed price is also bounded. Combining these with the convergence rate of the least squares estimates of α and β, we conclude in the proof of Theorem 2 that the T-period revenue loss of the slow-moving pricing policy is at most of order $\sqrt{T}$; see the proof of Theorem 2 in Online Appendix A for further details. The constant c_3 in Theorem 2 depends on $\underline{p}$, $\overline{p}$, Θ, ζ, and c_0, and is in the order of $1/[(1-\zeta)c_0]^2$.

A remarkable feature of the slow-moving pricing policy is that neither the parameters η^+ and η^-, which measure the impact of reference prices on the demand, nor the exponential smoothing parameter ζ in the reference-price updating mechanism need to be

Figure 4. (Color online) Reference Prices and Estimates Under the Slow-Moving Pricing Policy



Notes. (a) Sample paths of the reference price sequence $\{r_t\}$ under the slowing-moving pricing policy $\pi = \mathcal{S}(\boldsymbol{n}, \boldsymbol{\delta}, \hat{a}_0, \hat{b}_0)$ in the setting of Figures 2(b) and 3(b), where $n_i = 2^i$ and $\delta_i = c_0 2^{-i/4}$ for all $i \in \mathbb{N}$, $c_0 = 0.1$, and $(\hat{a}_0, \hat{b}_0) = (2, 1)$. (b) Sample paths of the sequence $\{\hat{\alpha}_t/(2\hat{\beta}_t)\}$ for the same policy and setting, in which $\hat{\alpha}_t/(2\hat{\beta}_t)$ denotes the latest estimate of the optimal fixed price $\alpha/(2\beta)$ in period t. The problem parameters are the same as in Figures 2(b) and 3(b). The value of $\alpha/(2\beta)$ equals 1.1 and is shown in the dotted line in each panel.

learned or estimated by the policy. This does not mean that these parameters can be ignored, as explained in detail in Section 3.1. Instead, the slow-moving policy diminishes the influence of η^+, η^-, and ζ by a smartly chosen design of prices: by keeping the price constant during periods of exponentially increasing length, the loss caused by not taking these parameters into account is dominated by the loss caused by having to learn α and β from sales data. Simply put, the judicious design of the slow-moving pricing policy ensures that statistical errors (caused by the demand model uncertainty) dominate model-misspecification errors (caused by ignoring parameters related to reference effects).

We note that the slow-moving pricing policy executes its experimentation and optimization steps repeatedly over episodes of geometrically increasing length $n_i = 2^i$. This idea of using episodes of geometrically increasing length is similar to the widely studied *doubling trick*, which enables designing an asymptotically optimal policy without requiring the knowledge of T (see Slivkins 2019, Lattimore and Szepesvári 2020). Likewise, the slow-moving pricing policy does not rely on the knowledge of T, but the construction of this policy differs from the standard doubling trick because the procedure used in each episode i is fine-tuned over time based on the values of $\hat{p}^*_{i-1}$ and δ_i. Similar constructions are used by Wang et al. (2014) and Besbes and Zeevi (2015) in pricing problems without reference effects.

The slow-moving pricing policy exhibits good performance both with and without reference effects. As shown in Figure 5, the regret of this policy is small regardless of the presence of reference effects. Comparing this with Figure 2, we observe that the slow-moving pricing policy is robust to the potential negative impact of reference effects on revenue performance, whereas policies that use frequent and large price changes are vulnerable to the same impact.

3.3. Extension to More General Reference-Price Processes

This section extends the preceding analysis to more general reference-price updating mechanisms that allow jumps in the value of the reference price and in the exponential smoothing coefficient. We formulate these jumps via a sequence of change-points $\{\tau_k \in \mathbb{N} \cup \{\infty\}, k = 1, 2, \dots\}$ such that $\tau_1 = 1$ and $\tau_k < \tau_{k+1}$ for all k satisfying $\tau_{k+1} < \infty$. Given the change-point sequence $\{\tau_k\}_{k\in\mathbb{N}}$, we generalize the reference-price updating mechanism by considering function sequences $\rho = \{\rho_t\}_{t\in\mathbb{N}}$ with $\rho_1 \in [\underline{p}, \overline{p}]$ and $\rho_t \in \mathcal{C}([\underline{p}, \overline{p}]^{t-1}, [\underline{p}, \overline{p}])$ for all $t \geq 2$ such that

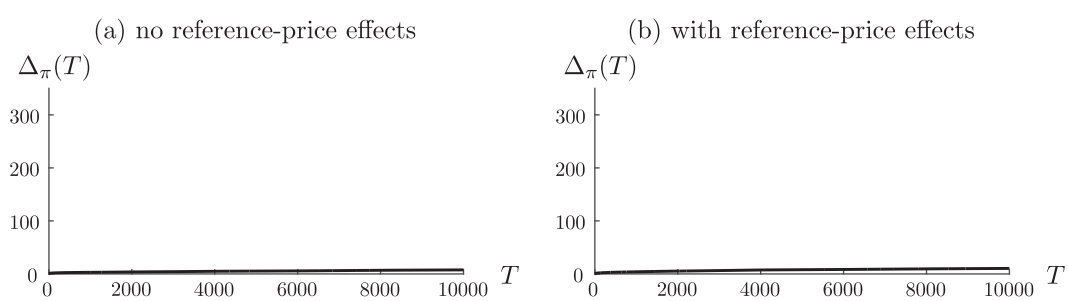
$$\rho_{t+1}(p_1, \dots, p_t) = \zeta_k \rho_t(p_1, \dots, p_{t-1}) + (1 - \zeta_k) p_t \quad \text{if} \quad \tau_k \leq t \leq \tau_{k+1} - 2, \tag{3.5}$$

for all $t \in \mathbb{N}, k \in \mathbb{N}$, and $\{p_t\}_{t\in\mathbb{N}} \in [\underline{p}, \overline{p}]^{\mathbb{N}}$, where $\zeta_k \in [0, 1)$ for all $k \in \mathbb{N}$. We emphasize that the seller knows neither $\{\zeta_k\}_{k\in\mathbb{N}}$ nor $\{\tau_k\}_{k\in\mathbb{N}}$. The recursion (3.5) imposes no restrictions on the reference-price formation if $t = \tau_{k+1} - 1$ for some $k \in \mathbb{N}$. This allows for jumps in the value of the reference-price process and in the exponential smoothing coefficient. These jumps occur in periods $\{\tau_k\}_{k\in\mathbb{N}}$. Note that, without any restrictions on the number of change points, it is possible to have a jump in every period, in which case the seller's regret would grow linearly in T. Consequently, to consider an asymptotic regime in which the seller can achieve the asymptotically optimal performance benchmark in the preceding section, suppose that we have the following bound on the number of change points: there is a finite and positive constant c such that

$$\sum_{\ell=s+1}^{t} \mathbb{I}\{\ell = \tau_k \text{ for some } k \in \mathbb{N}\} = \sum_{\ell=s+1}^{t} \sum_{k=1}^{\infty} \mathbb{I}\{\ell = \tau_k\} \leq c\sqrt{t - s} \tag{3.6}$$

for all $s, t \in \mathbb{N}$ satisfying $s < t$. That is, the number of change points within N consecutive periods is in the order of $\sqrt{N}$. To formally express the generalized

Figure 5. Regret of the Slow-Moving Pricing Policy



Notes. The graphs show the T-period regret of the slowing-moving pricing policy $\pi = \mathcal{S}(\boldsymbol{n}, \boldsymbol{\delta}, \hat{a}_0, \hat{b}_0)$ in the settings of Figure 2, in which $n_i = 2^i$ and $\delta_i = c_0 2^{-i/4}$ for all $i \in \mathbb{N}$, $c_0 = 0.1$, and $(\hat{a}_0, \hat{b}_0) = (2, 1)$. (a) is generated in the setting with no reference effects and (b) in the setting with reference effects. The problem parameters are the same as in Figure 2.

reference-price updating mechanism described, we define

$$\mathcal{T} = \left\{ \begin{array}{l} \{\tau_k \in \mathbb{N} \cup \{\infty\}\}_{k\in\mathbb{N}} : \tau_1 = 1, \tau_k < \tau_{k+1} \\ \quad \text{for all } k \text{ satisfying } \tau_{k+1} < \infty, \text{ and} \\ \sum_{\ell=s+1}^{t} \sum_{k=1}^{\infty} \mathbb{I}\{\ell = \tau_k\} \le c\sqrt{t-s} \\ \quad \text{for all } s,t \in \mathbb{N} \text{ satisfying } s < t \end{array} \right\},$$

and let

$$\rho \in \mathcal{R}_{\mathrm{G}} = \cup_{\zeta\in[0,1)^{\mathbb{N}}} \cup_{\tau\in\mathcal{T}} \mathcal{R}_{\zeta,\tau},$$

where

$$\mathcal{R}_{\zeta,\tau} = \left\{ \begin{array}{l} \{\rho_t\}_{t\in\mathbb{N}} : \rho_{t+1}(p_1,\ldots,p_t) \\ \quad = \zeta_k \rho_t(p_1,\ldots,p_{t-1}) + (1-\zeta_k)p_t \\ \quad \text{if } \tau_k \le t \le \tau_{k+1} - 2 \\ \text{for all } t \in \mathbb{N}, k \in \mathbb{N}, \text{ and } \{p_t\}_{t\in\mathbb{N}} \in [\underline{p}, \overline{p}]^{\mathbb{N}}, \\ \text{and } \rho_1 \in [\underline{p}, \overline{p}] \end{array} \right\}$$

for all $\zeta = \{\zeta_k\}_{k\in\mathbb{N}} \in [0,1)^{\mathbb{N}}$ and $\tau = \{\tau_k\}_{k\in\mathbb{N}} \in \mathcal{T}$.

To establish the asymptotic optimality of the slow-moving pricing policy in the preceding setting, we prove a generalized counterpart of Theorem 2. As explained earlier, the proof of Theorem 2 is based on Theorem 1 and Proposition 1. In fact, the tightness of the bounds in Theorem 1 and Proposition 1 allows us to characterize a broader class of reference-price updating mechanisms such that the conclusion of Theorem 2 still holds. We formally express this result in the following proposition.

Proposition 2 (Generalized Conditions for Asymptotic Optimality). *Suppose that $\rho = \{\rho_t\}_{t\in\mathbb{N}}$ satisfies $\rho_1 \in [\underline{p}, \overline{p}]$ and $\rho_t \in \mathcal{C}([\underline{p}, \overline{p}]^{t-1})$ for all $t \ge 2$ and that there exist finite and positive constants c_4 and c_5 such that, for all $T \in \mathbb{N}$ and $\theta = (\alpha, \beta, \eta^+, \eta^-) \in \Theta$,*

$$\Delta_{\text{fixed}}\left(\frac{\alpha}{2\beta}; T, \rho, \theta\right) \le c_4\sqrt{T}, \tag{3.7}$$

and for all $s,t \in \mathbb{N}$ satisfying $s < t$ and all $p_1,\ldots,p_s,p \in [\underline{p}, \overline{p}]$,

$$\sum_{k=s+1}^{t} \left| \rho_k\left(p_1,\ldots,p_s, \underbrace{p,\ldots,p}_{k-s-1 \text{ times}}\right) - p \right| \le c_5\sqrt{t-s}. \tag{3.8}$$

Moreover, let $\theta \in \Theta$, and $\pi = \mathcal{S}(\boldsymbol{n}, \boldsymbol{\delta}, \hat{a}_0, \hat{b}_0)$, where $n_i = 2^i$ and $\delta_i = c_0 2^{-i/4}$ for all $i \in \mathbb{N}$, $c_0 \in (0, 2^{-3/4}(\overline{p} - \underline{p}))$, and $(\hat{a}_0, \hat{b}_0) \in [\alpha_{\min}, \alpha_{\max}] \times [\beta_{\min}, \beta_{\max}]$. Then, there exists a finite and positive constant c_6 such that

$$\Delta_\pi(T, \rho, \theta) \le c_6\sqrt{T} \text{ for all } T \in \mathbb{N}.$$

Based on the preceding proposition, the following result extends Theorem 2 to the generalized reference-price updating mechanisms in $\mathcal{R}_{\mathrm{G}}$.

Theorem 3 (Asymptotic Optimality of the Slow-Moving Pricing Policy). *Let $\rho \in \mathcal{R}_{\mathrm{G}}$, $\theta \in \Theta$, and $\pi = \mathcal{S}(\boldsymbol{n}, \boldsymbol{\delta}, \hat{a}_0, \hat{b}_0)$, where $n_i = 2^i$ and $\delta_i = c_0 2^{-i/4}$ for all $i \in \mathbb{N}$, $c_0 \in (0, 2^{-3/4}(\overline{p} - \underline{p}))$, and $(\hat{a}_0, \hat{b}_0) \in [\alpha_{\min}, \alpha_{\max}] \times [\beta_{\min}, \beta_{\max}]$. Then,*

$$\Delta_\pi(T, \rho, \theta) \le c_6\sqrt{T} \text{ for all } T \in \mathbb{N},$$

where c_6 is as given in Proposition 2.

The constant c_6 depends on $\underline{p}$, $\overline{p}$, Θ, ζ, c_0, c_4, and c_5, and is in the order of $c_4 + c_5 + c_5^2/c_0^2$.

The class of mechanisms in (3.5) and (3.6) determine the extent of environments in which the asymptotic optimality benchmark (i.e., regret of order $\sqrt{T}$) can be achieved. If (3.6) does not necessarily hold, then the reference-price sequence $\{r_t\}$ can include more than $T^{\frac{1}{2}+\varepsilon}$ jumps within T periods, where $\varepsilon > 0$. Because the seller does not observe the reference-price sequence, the seller has to incur a revenue loss of order one after each jump, and the regret of any given policy is at least of order $T^{\frac{1}{2}+\varepsilon}$ in this case.

4. Incumbent-Price Setting with Reference Effects

In this section, we consider the case of a fixed reference price. This case is motivated by settings in which the seller has been charging an incumbent price $\tilde{p}_0 \in (\underline{p}, \overline{p})$ for a sufficiently long period of time and has thus learned the expected demand under that price. Because the incumbent price $\tilde{p}_0$ has been charged for a long time, customers perceive $\tilde{p}_0$ as a fixed reference price. Accordingly, we call this setting the *incumbent-price formulation with reference effects*. It is worth noting that the seller can efficiently identify whether the reference price is fixed at $\tilde{p}_0$, and we provide an asymptotically optimal statistical test for this purpose; see Online Appendix E.

4.1. Notation for the Incumbent-Price Formulation

Let $r \in (\underline{p}, \overline{p})$ be the fixed reference price; that is, $r_t = r = \tilde{p}_0$ for all t. More formally, we let

$$\mathcal{R}_{\mathrm{F}} = \{\{\rho_t\}_{t\in\mathbb{N}} : \rho_t = r \in (\underline{p}, \overline{p}) \text{ for all } t \ge 1\},$$

and assume that $\rho \in \mathcal{R}_{\mathrm{F}}$. Denote by $x := p - r$ the deviation of the selling price p from the incumbent price r. Then, the expected demand as a function of x can be expressed as follows:

$$\tilde{d} : [\underline{p} - r, \overline{p} - r] \to \mathbb{R}_+,$$
$$\tilde{d}(x) = \begin{cases} \tilde{d}_0 - \beta^+ x & \text{if } x < 0, \\ \tilde{d}_0 - \beta^- x & \text{if } x \ge 0, \end{cases} \tag{4.1}$$

where $\tilde{d}_0 = d(r, r) = \alpha - \beta r$, $\beta^+ = \beta + \eta^+$, and $\beta^- = \beta + \eta^-$. Based on (2.1) and (4.1), the demand realization in period t is given by

$$d_t = \tilde{d}(x_t) + \epsilon_t, \tag{4.2}$$

for all $t = 1, \ldots, T$, where $x_t = p_t - r$ is the deviation from the incumbent price in period t.

Recalling that $(\beta, \eta^+, \eta^-) \in [\beta_{\min}, \beta_{\max}] \times [\eta^+_{\min}, \eta^+_{\max}] \times [\eta^-_{\min}, \eta^-_{\max}]$, we deduce that $\beta^+ \in [\beta^+_{\min}, \beta^+_{\max}]$ and $\beta^- \in [\beta^-_{\min}, \beta^-_{\max}]$, where $\beta^+_{\min} = \beta_{\min} + \eta^+_{\min}$, $\beta^+_{\max} = \beta_{\max} + \eta^+_{\max}$, $\beta^-_{\min} = \beta_{\min} + \eta^-_{\min}$, and $\beta^-_{\max} = \beta_{\max} + \eta^-_{\max}$. Furthermore, because we consider loss-averse customers, we have $\eta^+ \le \eta^-$, which implies that $\beta^+ \le \beta^-$. Consequently, (β^+, β^-) lies in the compact set B defined as follows:

$$B := \{(\beta^+, \beta^-) \in [\beta^+_{\min}, \beta^+_{\max}] \times [\beta^-_{\min}, \beta^-_{\max}] : \beta^+ \le \beta^-\}. \tag{4.3}$$

As before, to highlight the dependence of the demand function on unknown parameters, we occasionally employ the notation $\tilde{d}(x, \beta^+, \beta^-)$ in place of $\tilde{d}(x)$. In accordance with this, the expected single-period revenue function is given by

$$\tilde{R}(x, \beta^+, \beta^-) = (r + x)\tilde{d}(x, \beta^+, \beta^-) = \begin{cases} (r + x)(\tilde{d}_0 - \beta^+ x) & \text{if } x < 0, \\ (r + x)(\tilde{d}_0 - \beta^- x) & \text{if } x \ge 0, \end{cases} \tag{4.4}$$

for all $x \in [\underline{p} - r, \overline{p} - r]$ and $(\beta^+, \beta^-) \in B$. Thus, by the first-order optimality conditions, the seller's full-information optimal price deviation that maximizes $\tilde{R}(x, \beta^+, \beta^-)$ is

$$\psi(\beta^+, \beta^-) = \begin{cases} -\dfrac{r}{2} + \dfrac{\tilde{d}_0}{2\beta^+} & \text{if } \dfrac{\tilde{d}_0}{r} < \beta^+, \\ 0 & \text{if } \beta^+ \le \dfrac{\tilde{d}_0}{r} \le \beta^-, \\ -\dfrac{r}{2} + \dfrac{\tilde{d}_0}{2\beta^-} & \text{if } \beta^- < \dfrac{\tilde{d}_0}{r}, \end{cases} \tag{4.5}$$

for all $(\beta^+, \beta^-) \in B$. Figure 6 illustrates the shape of the revenue function $\tilde{R}(x, \beta^+, \beta^-)$ and its optimizer $\psi(\beta^+, \beta^-)$ for different values of β^+ and β^-.

Based on these expressions of the revenue function and the optimal price deviation, we let $\tilde{\Delta}_\pi(T, \rho, \beta^+, \beta^-)$ denote the regret in the incumbent-price formulation. That is, in the incumbent-price setting, we have $\Delta_\pi(T, \rho, \theta) = \tilde{\Delta}_\pi(T, \rho, \beta^+, \beta^-) = T\tilde{R}^*(\beta^+, \beta^-) - \mathbb{E}_\pi[\sum_{t=1}^T \tilde{R}(x_t, \beta^+, \beta^-)]$, where $\tilde{R}^*(\beta^+, \beta^-) = \sup_{x \in [\underline{p} - r, \overline{p} - r]}\{\tilde{R}(x, \beta^+, \beta^-)\}$. We also define the worst-case regret of a policy π in the incumbent-price setting as $\tilde{\Delta}_\pi(T, \rho) = \sup_{(\beta^+, \beta^-) \in B}\{\tilde{\Delta}_\pi(T, \rho, \beta^+, \beta^-)\}$.

4.2. A Lower Bound on Regret

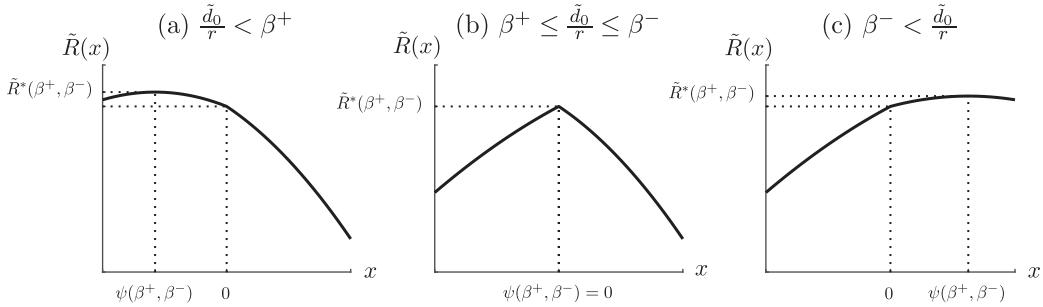
The next result characterizes the best achievable regret performance in the incumbent-price formulation with reference effects.

Proposition 3 (Lower Bound on Regret in the Incumbent-Price Problem). *Let $\rho \in \mathcal{R}_F$. Assume that the demand function is given by the incumbent-price formulation in (4.1), and that $\epsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, where $\sigma > 0$. Then, there exists a finite and positive constant $\tilde{c}_1$ such that, under any admissible policy π,*

$$\tilde{\Delta}_\pi(T, \rho) \ge \tilde{c}_1 \sqrt{T} \quad \textit{for all } T \in \mathbb{N}.$$

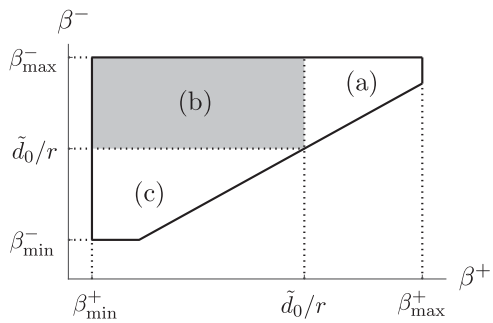
Proposition 3 shows that, in the incumbent-price setting with reference effects, the worst-case regret of any given policy is at least of order $\sqrt{T}$. This stands in contrast with earlier findings on the incumbent-price formulation in the absence of reference effects. For instance, Keskin and Zeevi (2014) show that it is possible to achieve a regret of order $\log T$ in the incumbent-price formulation with no reference effects (see Keskin and Zeevi 2014, section 4). The reason for this difference is that, under reference effects, the optimal selling price equals the incumbent price for a large set of demand parameter values (see the shaded area in Figure 7), and it is exceedingly difficult to collect information about the unknown demand parameters by charging prices close to the incumbent price. As a result,

Figure 6. (Color online) Revenue Function Examples in the Incumbent-Price Setting



Notes. (a)–(c) display the revenue function $\tilde{R}(x)$ for the three different cases in (4.5). If $\tilde{d}_0/r < \beta^+$, then the optimal price deviation $\psi(\beta^+, \beta^-)$ is the unique stationary point on the left of $x = 0$; see (a). If $\beta^- < \tilde{d}_0/r$, then $\psi(\beta^+, \beta^-)$ is the unique stationary point on the right of $x = 0$; see (c). If $\beta^+ \le \tilde{d}_0/r \le \beta^-$, then $\tilde{R}(x)$ is not differentiable at its optimum, and $\psi(\beta^+, \beta^-) = 0$; see (b). Problem parameters are $r = 1$ and $\tilde{d}_0 = 1$ on all three panels, whereas $(\beta^+, \beta^-) = (1.3, 1.5)$ on (a), $(\beta^+, \beta^-) = (0.6, 1.5)$ on (b), and $(\beta^+, \beta^-) = (0.6, 0.8)$ on (c).

Figure 7. (Color online) Demand Parameter Regions in the Incumbent-Price Setting



Notes. The graph displays the feasible set of estimates, namely, $B = \{(\beta^+, \beta^-) \in [\beta^+_{\min}, \beta^+_{\max}] \times [\beta^-_{\min}, \beta^-_{\max}] : \beta^+ \le \beta^-\}$. In region (a), the optimal price deviation is negative; that is, $\psi(\beta^+, \beta^-) < 0$, whereas in region (c), $\psi(\beta^+, \beta^-) > 0$. Finally, in region (b), $\psi(\beta^+, \beta^-) = 0$.

in the presence of reference effects, the seller has to incur a larger revenue loss because of demand model uncertainty.

4.3. Designing an Asymptotically Optimal Pricing Policy

Having characterized the best achievable performance in the incumbent-price problem with reference effects, we now design and analyze a policy that achieves this performance benchmark. Note that the slow-moving policy introduced in Section 3 is not optimal in the setting of this section: a fixed reference price does not converge to a selling price that is kept constant for a long amount of time, and thus, Proposition 1, which is essential for proving the good performance of the slow-moving pricing policy, no longer holds. This is why we construct a new policy in this section. Because there are a smaller number of unknown demand parameters in the incumbent-price formulation, the estimation procedure can be simplified as follows. Given the price deviations $(x_1, \dots, x_t)$ and the demand realizations $(d_1, \dots, d_t)$ observed through the end of period t, the least squares estimators of β^+ and β^- are given by

$$\hat{\beta}^+_t \in \arg\min_{\beta^+} \left\{ \sum_{s=1}^{t} (\tilde{d}_0 - \beta^+ x_s - d_s)^2 \, \mathbb{I}\{x_s < 0\} \right\}, \tag{4.6a}$$

$$\hat{\beta}^-_t \in \arg\min_{\beta^-} \left\{ \sum_{s=1}^{t} (\tilde{d}_0 - \beta^- x_s - d_s)^2 \, \mathbb{I}\{x_s \ge 0\} \right\}, \tag{4.6b}$$

respectively, where $\mathbb{I}\{\cdot\}$ is the indicator function (that is, given condition A, $\mathbb{I}\{A\} = 1$ if A holds and zero otherwise). Note that $\hat{\beta}^+_t$ and $\hat{\beta}^-_t$ are well defined as long as the seller makes at least one negative and one nonnegative price deviation relative to the incumbent price: if $\sum_{s=1}^{t} \mathbb{I}\{x_s < 0\} > 0$ and $\sum_{s=1}^{t} \mathbb{I}\{x_s \ge 0\} > 0$, then $\hat{\beta}^+_t$ and $\hat{\beta}^-_t$ are computed via the unique solution to (4.6a) and (4.6b). To initialize the estimation in the beginning of the time horizon, we let $(\hat{b}^+_0, \hat{b}^-_0) \in B$ be the seller's initial estimate vector. (In practice, these initial estimates can depend on historical data and/or subjective predictions.)

Because (β^+, β^-) lies in the compact set $B = \{(\beta^+, \beta^-) \in [\beta^+_{\min}, \beta^+_{\max}] \times [\beta^-_{\min}, \beta^-_{\max}] : \beta^+ \le \beta^-\}$, the unconstrained estimates $\hat{\beta}^+_t$ and $\hat{\beta}^-_t$ can be truncated to improve accuracy. Note that the initial estimate vector $(\hat{b}^+_0, \hat{b}^-_0)$ is already in B, and in any period $t \ge 1$, the truncated estimates are computed as follows. Let $(\hat{b}^+_{t-1}, \hat{b}^-_{t-1}) \in B$ be the truncated estimates computed in period $t-1$. Because x_t is either negative or nonnegative, we deduce from (4.6a) and (4.6b) that only one of the estimators $\hat{\beta}^+_t$ and $\hat{\beta}^-_t$ is updated in period t. Therefore,

$$(\hat{b}^+_t, \hat{b}^-_t) = \begin{cases} \left(\mathcal{P}_{[\beta^+_{\min}, \hat{b}^-_{t-1}]}\{\hat{\beta}^+_t\}, \hat{b}^-_{t-1} \right) & \text{if } x_t < 0, \\ \left(\hat{b}^+_{t-1}, \mathcal{P}_{[\hat{b}^+_{t-1}, \beta^-_{\max}]}\{\hat{\beta}^-_t\} \right) & \text{if } x_t \ge 0, \end{cases} \tag{4.7}$$

where $\mathcal{P}_{[\ell,u]}\{x\} = (x \vee \ell) \wedge u$ for $\ell, u, x \in \mathbb{R}$. Note that regardless of the sign of x_t, $(\hat{b}^+_t, \hat{b}^-_t) \in B$.

Based on this estimation procedure, we propose the following policy.

Quick Learning Policy $\mathcal{Q}(\boldsymbol{n}, \delta, \hat{b}^+_0, \hat{b}^-_0)$

Initialization. Let $\boldsymbol{n} = \{n_i \in \mathbb{N}, i = 1, 2, \dots\}$ such that $n_i \ge 2$ for all $i \ge 2$, $\delta \in (0, (\overline{p} - r) \wedge (r - \underline{p}))$, and $(\hat{b}^+_0, \hat{b}^-_0) \in B$. Define

$$\mathcal{T}^- = \left\{ \sum_{j=1}^{i} n_j \in \mathbb{N}, \, i = 1, 2, \dots \right\} \text{ and}$$

$$\mathcal{T}^+ = \left\{ \sum_{j=1}^{i} n_j + 1 \in \mathbb{N}, \, i = 1, 2, \dots \right\}.$$

Fix $\hat{p}^*_0 = r + \psi(\hat{b}^+_0, \hat{b}^-_0)$, and for all $t = 1, 2, \dots$ satisfying $t \le T$, do the following:

Price Experimentation. Choose

$$p_t = \begin{cases} r - \delta & \text{if } t \in \mathcal{T}^-, \\ r + \delta & \text{if } t \in \mathcal{T}^+, \\ \hat{p}^*_{t-1} & \text{otherwise.} \end{cases}$$

Demand Estimation. Compute $\hat{b}^+_t$ and $\hat{b}^-_t$ via (4.7).

Price Optimization. Set $\hat{p}^*_t := r + \psi(\hat{b}^+_t, \hat{b}^-_t)$ via (4.5).

The preceding policy is called the *quick learning policy* with parameters $(\boldsymbol{n}, \delta, \hat{b}^+_0, \hat{b}^-_0)$ and denoted by $\mathcal{Q}(\boldsymbol{n}, \delta, \hat{b}^+_0, \hat{b}^-_0)$. This policy, as does its counterpart in the preceding section, divides the time horizon into consecutive episodes indexed by $i \in \mathbb{N}$. At the beginning of each

episode, $\mathcal{Q}(\boldsymbol{n},\delta,\hat{b}_0^+,\hat{b}_0^-)$ charges two test prices, namely, $r-\delta$ and $r+\delta$, to accelerate the seller's learning. Then, until the end of the episode, said policy keeps charging an estimate of the optimal price, $\hat{p}_{t-1}^*$, that is based on all available data. The episode lengths $\boldsymbol{n}=\{n_i, i=1, 2,\dots\}$, the test-price perturbation δ, and the initial estimates $(\hat{b}_0^+, \hat{b}_0^-)$ are input parameters of the policy.

Our following result provides a performance guarantee for the quick learning policy.

Theorem 4 (Asymptotic Optimality of the Quick Learning Policy). *Let $\rho\in\mathcal{R}_F$, $(\beta^+,\beta^-)\in B$, and $\pi=\mathcal{Q}(\boldsymbol{n},\delta,\hat{b}_0^+,\hat{b}_0^-)$, where $n_i = i$ for all $i\in\mathbb{N}$, $\delta\in(0,(\overline{p}-r)\wedge(r-\underline{p}))$, and $(\hat{b}_0^+,\hat{b}_0^-)\in B$. Assume that the demand function is given by the incumbent-price formulation in (4.1). Then, there exists a finite and positive constant $\tilde{c}_2$ such that*

$$\tilde{\Delta}_\pi(T,\rho,\beta^+,\beta^-)\le\tilde{c}_2\sqrt{T}\ \text{ for all } T\in\mathbb{N}.$$

Theorem 4 shows that the quick learning policy is asymptotically optimal as it achieves the smallest possible growth rate of regret, which is given in Proposition 3.

It is worth noting that Wang (2012) considers a pricing-and-learning problem in which the revenue function has a kink at the optimal price, similar to Figure 6(b) (Wang 2012). A key difference between our setting and that of Wang (2012) is that Wang (2012) requires the knowledge that the demand function has a kink at the optimal price. By contrast, in our setting, the price at which the revenue function is nondifferentiable may or may not coincide with the optimal price as visualized in Figure 6, and the quick learning policy does not need to know this in advance.

4.4. Incomplete Learning Under Reference Effects

A key question about learning and earning under reference effects is concerned with the performance of *passive learning*. The policies in Sections 3.2 and 4.3 employ forced exploration in the form of price experimentation, which enables them to learn the demand model at an asymptotically optimal rate. By contrast, passive learning policies make no forced-exploration attempt. In each sales opportunity, these policies charge the myopically optimal price based on the most recent demand estimation and passively update the estimates based on the observed demand response. In our setting, this approach can be formulated as a policy called the *certainty-equivalence pricing policy*, which charges the myopically optimal price in each period. The certainty-equivalence pricing policy with initial estimates $(\hat{b}_0^+,\hat{b}_0^-)$, hereafter abbreviated as $\mathrm{CE}(\hat{b}_0^+,\hat{b}_0^-)$, uses the price $p_t=r+\psi(\hat{b}_{t-1}^+,\hat{b}_{t-1}^-)$ in period $t=1,2,\dots$ Note that $\mathrm{CE}(\hat{b}_0^+,\hat{b}_0^-)$ is a special case of the policy $\mathcal{Q}(\boldsymbol{n},\delta,\hat{b}_0^+,\hat{b}_0^-)$ in Section 4.3 in the sense that it conducts no price experiments throughout the time horizon (i.e., the first experimentation period n_1 is greater than T).

Because of their simplicity, passive learning policies are commonly used in pricing practice (Harrison et al. 2012, section 1). But these policies typically suffer from a complication called *incomplete learning*, which occurs when their estimates converge to incorrect values. In the absence of reference effects, the antecedent literature on dynamic pricing with demand learning establishes that, under a passive learning policy, incomplete learning occurs with strictly positive probability (Harrison et al. 2012, den Boer and Zwart 2014, Keskin and Zeevi 2018). We show that reference effects severely exacerbate the incomplete learning complication of passive learning policies. To be precise, we derive a stronger result that establishes that, in the incumbent-price formulation with reference effects, a passive learning policy can exhibit incomplete learning with probability one.

Theorem 5 (Strong Incomplete Learning Under Reference Effects). *Let $\rho\in\mathcal{R}_F$, $(\beta^+,\beta^-)\in B$, and $\pi=\mathrm{CE}(\hat{b}_0^+,\hat{b}_0^-)$, where $(\hat{b}_0^+,\hat{b}_0^-)\in B$. Assume that the demand function is given by the incumbent-price formulation in (4.1). If $\psi(\beta^+,\beta^-)<0$ and $\psi(\hat{b}_0^+,\hat{b}_0^-)\ge 0$, then*

$$\mathbb{P}_\pi\{(\hat{b}_t^+,\hat{b}_t^-)\to(\beta^+,\beta^-)\}=0.$$

Similarly, if $\psi(\beta^+,\beta^-)>0$ and $\psi(\hat{b}_0^+,\hat{b}_0^-)\le 0$, then

$$\mathbb{P}_\pi\{(\hat{b}_t^+,\hat{b}_t^-)\to(\beta^+,\beta^-)\}=0.$$

Remark 2. In the incumbent-price problem without reference effects, certainty-equivalence pricing is known to suffer from incomplete learning with some positive probability less than one (see, e.g., Keskin and Zeevi 2018). Theorem 5 shows that the presence of reference effects makes certainty-equivalence pricing exhibit incomplete learning with probability one in this setting. This is how reference effects worsen the incomplete learning complication of certainty-equivalence pricing.

According to Theorem 5, if the underlying optimal price and the initial price of the certainty-equivalence policy are on different sides of the reference price, then the certainty-equivalence policy almost surely fails to identify the underlying model parameters. Because this is an almost sure statement, it indicates that the presence of reference effects intensifies the incomplete learning phenomenon. The essential reason for this is that reference effects strongly encourage the

seller to charge an uninformative price over a large set of values for demand parameter estimates (see Figure 6(b) and the shaded region in Figure 7).

An important consequence of the incomplete learning result in Theorem 5 is that the seller's regret grows linearly in T, which is the highest possible growth rate of regret in our setting.

Corollary 1 (Loss Resulting from Incomplete Learning). *Let $\rho \in \mathcal{R}_F$, $(\beta^+, \beta^-) \in B$, and $\pi = \mathrm{CE}(\hat{b}_0^+, \hat{b}_0^-)$, where $(\hat{b}_0^+, \hat{b}_0^-) \in B$. Assume that the demand function is given by the incumbent-price formulation in (4.1). If $\psi(\beta^+, \beta^-) < 0$ and $\psi(\hat{b}_0^+, \hat{b}_0^-) \geq 0$ (or similarly, if $\psi(\beta^+, \beta^-) > 0$ and $\psi(\hat{b}_0^+, \hat{b}_0^-) \leq 0$), then there exists a finite and positive constant $\tilde{c}_3$ such that*

$$\tilde{\Delta}_\pi(T, \rho, \beta^+, \beta^-) \geq \tilde{c}_3 T \quad \text{for all} \quad T \in \mathbb{N}.$$

Remark 3. The preceding results demonstrate the extent and ramifications of incomplete learning in the incumbent-price setting. Outside this setting, further complications arise in the implementation of passive learning. For instance, in the case of time-varying reference prices studied in Section 3, the seller neither observes the reference price r_t nor knows the exponential smoothing coefficient ζ. Without the knowledge of these variables, the seller cannot construct a well-defined myopically optimal pricing function and may have to use a misspecified demand model with a smaller number of parameters. Therefore, on top of incomplete learning, there is an identifiability problem for passive learning under time-varying reference prices. Theorem 5 and Corollary 1 show that, even in the absence of such identifiability complications, reference effects severely worsen the performance of passive learning policies.

5. Extension to Gain-Seeking Customer Behavior

In this section, we extend our analysis to the case of *gain-seeking* customers. In the notation of Section 2, this means that $\eta^+ > \eta^-$.

As before, the seller's performance under demand model uncertainty is based on the optimal price path under full information. In contrast to the case of loss aversion, in which the full-information optimal price path is asymptotically constant, gain-seeking customers can induce a wide variety of price paths: as shown by Hu et al. (2016), the full-information optimal price path for gain-seeking customers is related to discrete dynamical systems defined by a discontinuous map that results in complex dynamics. Because of this complexity, Hu et al. (2016) do not derive explicit expressions for the full-information optimal pricing policy except for a setting in which $\eta^+ > 0$, $\eta^- = 0$, and the exponential smoothing coefficient $\zeta = 0$. Under these conditions, Hu et al. (2016) show that a cyclic price-skimming strategy, which repeatedly uses a decreasing price sequence $\{p_1, \ldots, p_m\}$, is optimal under full information. To the best of our knowledge, Hu et al. (2016) present the state of the art in characterization of full-information optimal pricing policies for gain-seeking customers. Because of this as well as the fact that demand model uncertainty adds a substantial layer of complexity on top of the full-information optimal price path, we use the preceding conditions considered by Hu et al. (2016) in our setting.

Our investigation of gain-seeking customer behavior under demand model uncertainty reveals a set of distinct results: in this case, the optimal growth rate of regret is parameter-dependent, and in particular, it depends on the cycle length of the best cyclic price-skimming policy under full information. In what follows, we study this phenomenon in detail.

As explained, we employ the aforementioned conditions in Hu et al. (2016) to characterize the full-information optimal pricing policy. For this purpose, we consider the setting in Section 2 but now with $\eta^- = 0$ and $\eta^+ > 0$. With slight abuse of notation, throughout this section, we write η instead of η^+, denote the vector of unknown demand parameters by $\theta = (\alpha, \beta, \eta)$, and suppose that θ resides in the following compact set:

$$\Theta := \{(\alpha, \beta, \eta) \in [\alpha_{\min}, \alpha_{\max}] \times [\beta_{\min}, \beta_{\max}] \times [\eta_{\min}, \eta_{\max}]\}, \tag{5.1}$$

where $\alpha_{\min}$, $\alpha_{\max}$, $\beta_{\min}$, $\beta_{\max}$, $\eta_{\min}$, $\eta_{\max}$ are known constants that satisfy $[\alpha_{\min}, \alpha_{\max}] \times [\beta_{\min}, \beta_{\max}] \times [\eta_{\min}, \eta_{\max}] \subset [0, \infty) \times (0, \infty) \times (0, \infty)$. To ensure that the underlying revenue function is concave, we assume that $\eta + \varepsilon_0 \leq 2\beta$ for all $\theta = (\alpha, \beta, \eta) \in \Theta$, where $\varepsilon_0 > 0$. Similar to Hu et al. (2016), we consider reference prices that are updated via exponential smoothing with coefficient $\zeta = 0$; that is, $\rho \in \mathcal{R}_0$, where $\mathcal{R}_0$ is as in (3.2).

We first analyze the optimal revenue under full information. To this end, let $\mathbf{S}_m$ denote the set of nonincreasing and nonnegative price sequences of length m; that is,

$$\mathbf{S}_m := \{(p_1, \ldots, p_m) \in \mathbb{R}^m : p_1 \geq \cdots \geq p_m \geq 0\}.$$

For $m \in \mathbb{N}$, $\theta = (\alpha, \beta, \eta) \in \Theta$, and $(p_1, \ldots, p_m) \in \mathbb{R}^m$, let

$$R_m(p_1, \ldots, p_m; \theta) := \frac{1}{m} \sum_{i=1}^{m} p_i(\alpha - \beta p_i + \eta(p_{i-1} - p_i)\mathbb{I}\{i \geq 2\}). \tag{5.2}$$

If the seller repeatedly uses a nonincreasing price sequence (i.e., there exists $m \in \mathbb{N}$ such that $(p_1, \ldots, p_m) \in \mathbf{S}_m$ and $p_t = p_{t-m}$ for all $t > m$), then $R_m(p_1, \ldots, p_m; \theta)$ corresponds to the expected per-period revenue.

Our first result is an explicit expression for the price sequence that maximizes $R_m(p_1, \ldots, p_m; \theta)$ for each positive integer m. In addition, we establish that the

optimal cycle length is between two and a natural number M independent of θ.

Proposition 4 (Optimal Price Cycle for Gain-Seeking Customers). *Let $m \in \mathbb{N}$ and $\theta = (\alpha, \beta, \eta) \in \Theta$. Then, the function $R_m(p_1, \ldots, p_m; \theta)$ is strictly concave in $(p_1, \ldots, p_m)$ with unique maximizer $\mathbf{p}^*(\theta, m)$ satisfying*

$$\mathbf{p}^*(\theta, m)^{\mathrm{T}} = (p_1^*(\theta, m), \ldots, p_m^*(\theta, m))^{\mathrm{T}} := -H_m^{-1}(\theta)\frac{\alpha}{m}\mathbf{1}_{m\times 1}, \tag{5.3}$$

which satisfies $\mathbf{p}^*(\theta, m) \in \mathbf{S}_m$, where $\mathbf{1}_{m\times 1}$ denotes an $m \times 1$ matrix with one at each entry, and

$$H_m(\theta) := \frac{1}{m}\begin{bmatrix} -2\beta & \eta & 0 & \cdots & 0 & 0 & 0 \\ \eta & -2(\beta+\eta) & \eta & \ddots & 0 & 0 & 0 \\ 0 & \eta & -2(\beta+\eta) & \ddots & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & \ddots & -2(\beta+\eta) & \eta & 0 \\ 0 & 0 & 0 & \ddots & \eta & -2(\beta+\eta) & \eta \\ 0 & 0 & 0 & \cdots & 0 & \eta & -2(\beta+\eta) \end{bmatrix} \tag{5.4}$$

is the $m \times m$ Hessian matrix of $R_m(p_1, \ldots, p_m; \theta)$ for all $(p_1, \ldots, p_m)$. Furthermore,

$$R_1(\mathbf{p}^*(\theta, 1); \theta) < R_2(\mathbf{p}^*(\theta, 2); \theta) \text{ for all } \theta \in \Theta, \tag{5.5}$$

and there exists $M \in \mathbb{N}$ such that

$$R_m(\mathbf{p}^*(\theta, m); \theta) < R_2(\mathbf{p}^*(\theta, 2); \theta) \quad \text{for all } m > M \text{ and } \theta \in \Theta. \tag{5.6}$$

To prove this proposition, we derive the first and second derivatives of the function $(p_1, \ldots, p_m) \longmapsto R_m(p_1, \ldots, p_m; \theta)$ for each $m \in \mathbb{N}$. From this, we deduce that said function is strictly concave and maximized by the unique solution to the first order conditions. To show that this solution is a nonincreasing sequence, that is, that $\mathbf{p}^*(\theta, m)$ lies in $\mathbf{S}_m$, we prove that, for any sequence of nonnegative prices $p_1, \ldots, p_m$, the term $\sum_{i=2}^n p_i(p_{i-1} - p_i)$ is maximized over permutations of the price sequence if $p_1, \ldots, p_m$ are sorted in nonincreasing order. The lower bound (5.5) on the optimal cycle length is proven by explicitly calculating the involved expressions, and the upper bound (5.6) is proven by showing that $R_m(\mathbf{p}^*(\theta, m); \theta)$ converges to the optimal single-period revenue as m grows large. The details of the proof are in Online Appendix C.

For ease of exposition, we consider the case in which the boundary prices $\underline{p}$ and $\overline{p}$ do not affect the maximization of $R_m(p_1, \ldots, p_m; \theta)$. Formally, we let $\underline{p} < \min\{p_m^*(\theta, m) : \theta \in \Theta, m \in \{1, \ldots, M\}\}$ and $\max\{p_1^*(\theta, m) : \theta \in \Theta, m \in \{1, \ldots, M\}\} < \overline{p}$. Note that these quantities are well defined because the expression in (5.3) is continuous in θ, and Θ is compact. Let $R_m^*(\theta)$ denote the maximum value of $R_m(p_1, \ldots, p_m; \theta)$, that is,

$$R_m^*(\theta) := \max\{R_m(p_1, \ldots, p_m; \theta) : (p_1, \ldots, p_m) \in \mathbf{S}_m\} \quad \text{for } m \in \{1, \ldots, M\} \text{ and } \theta \in \Theta,$$

and let $m^*(\theta)$ be the optimal cycle length (in which ties are broken favoring the smallest cycle length):

$$m^*(\theta) := \min\{m \in \{1, \ldots, M\} : R_m^*(\theta) = \sup\{R_n^*(\theta) : n \in \mathbb{N}\}\} \quad \text{for } \theta \in \Theta.$$

Given T, ρ, and θ, let $\Delta_{\text{cycle}}(T, \rho, \theta)$ be the cumulative loss resulting from using the optimal nonincreasing price sequence in a cyclic fashion as opposed to the best dynamic sequence of prices under full information:

$$\Delta_{\text{cycle}}(T, \rho, \theta) = \Pi^*(T, \rho, \theta) - TR_{m^*(\theta)}^*(\theta). \tag{5.7}$$

The next result shows that repeatedly using the optimal nonincreasing price cycle is asymptotically optimal in our setting.

Proposition 5 (Loss Under the Best Cyclic Price-Skimming Policy). *Let $\rho \in \mathcal{R}_0$ and $\theta \in \Theta$. Then, there exists a finite and positive constant $\check{c}_1$ such that*

$$\Delta_{\text{cycle}}(T, \rho, \theta) \le \check{c}_1 \text{ for all } T \in \mathbb{N}.$$

We prove this result by observing that any price sequence $\{p_1, \ldots, p_T\}$ consists of consecutive nonincreasing price sequences. The expected per-period revenue obtained with a nonincreasing price sequence is bounded above by $R_{m^*(\theta)}^*$. By summing the expected revenue over all consecutive decreasing price sequences and taking the supremum over all $p_1, \ldots, p_T$, we obtain the statement of the proposition; see Online Appendix C for details.

We now construct a pricing policy for our setting of gain-seeking customers and demand model uncertainty. Given an admissible policy π, let

$$\mathcal{J}_t^\pi := \sum_{s=1}^t \begin{bmatrix} 1 \\ -p_s \\ (p_{s-1} - p_s)^+ \end{bmatrix}\begin{bmatrix} 1 \\ -p_s \\ (p_{s-1} - p_s)^+ \end{bmatrix}^{\mathrm{T}} \quad \text{for} \quad t = 1, 2, \ldots,$$

where $p_1, \ldots, p_T$ satisfy (2.6) and $p_0 = r_1$ is the initial reference price. Furthermore, let $\mathcal{J}_0^\pi$ be a 3×3 matrix with zero entries, and for any given matrix A, denote the smallest eigenvalue of A by $\lambda_{\min}(A)$. Based on this construction, we propose the following pricing policy for the case of gain-seeking customer behavior.

Gain-Seeking Customer Pricing Policy $\mathcal{G}$

Initialization. Let $\tilde{p}_1 = \overline{p}$, $\tilde{p}_2 = \frac{2}{3}\underline{p} + \frac{1}{3}\overline{p}$, $\tilde{p}_3 = \frac{1}{3}\underline{p} + \frac{2}{3}\overline{p}$, and $\tilde{p}_4 = \underline{p}$. Fix $\hat{m}_{-1} = 0$, $\hat{p} = \underline{p}$, $\tau_{-1} = 0$, and for all $c = 0, 1, \ldots,$ do the following:

Price Experimentation.

a. If $\lambda_{\min}(\mathcal{J}_{\tau_{c-1}}^{\mathcal{G}}) \le \sqrt{\tau_{c-1}}$, then set $\tau_c = \tau_{c-1} + c\hat{m}_{c-1} + 4$, and choose

$$p_t = \hat{p}(1 + i_{c-1}(t)) \quad \text{for } t = \tau_{c-1} + 1, \ldots, \tau_{c-1} + c\hat{m}_{c-1},$$

where $i_{c-1}(t)$ is the smallest nonnegative integer such that $i_{c-1}(t) \equiv t - \tau_{c-1} - 1 \pmod{\hat{m}_{c-1}}$, that is, $t - \tau_{c-1} - 1 - i_{c-1}(t)$ is an integer multiple of $\hat{m}_{c-1}$. Moreover, choose

$$p_t = \tilde{p}_j \quad \text{for} \quad t = \tau_{c-1} + c\hat{m}_{c-1} + j \quad \text{and} \quad j = 1,2,3,4.$$

b. If $\lambda_{\min}(\mathcal{J}^{\mathcal{G}}_{\tau_{c-1}}) > \sqrt{\tau_{c-1}}$, then set $\tau_c = \tau_{c-1} + c\hat{m}_{c-1}$, and choose

$$p_t = \hat{p}(1 + i_{c-1}(t)) \quad \text{for} \quad t = \tau_{c-1} + 1, \ldots, \tau_{c-1} + c\hat{m}_{c-1},$$

where $i_{c-1}(t)$ is the smallest nonnegative integer such that $i_{c-1}(t) \equiv t - \tau_{c-1} - 1 \pmod{\hat{m}_{c-1}}$.

Demand Estimation. Compute

$$(\hat{\alpha}_c, \hat{\beta}_c, \hat{\eta}_c) \in \arg\min_{(\alpha,\beta,\eta)} \left\{ \sum_{s=1}^{\tau_c} (\alpha - \beta p_s + \eta(p_{s-1} - p_s)^+ - d_s)^2 \right\},$$

and let $\hat{a}_c := \mathcal{P}_{[\alpha_{\min},\alpha_{\max}]}\{\hat{\alpha}_c\}$, $\hat{b}_c := \mathcal{P}_{[\beta_{\min},\beta_{\max}]}\{\hat{\beta}_c\}$, $\hat{h}_c := \mathcal{P}_{[\eta_{\min},\eta_{\max}]}\{\hat{\eta}_c\}$, where $\mathcal{P}_{[\ell,u]}\{x\} := (x \vee \ell) \wedge u$ for $\ell, u, x \in \mathbb{R}$. Moreover, let $\hat{\theta}_c := (\hat{a}_c, \hat{b}_c, \hat{h}_c)$.

Price Optimization. Set $\hat{m}_c := m^*(\hat{\theta}_c)$ and

$$(\hat{p}(1), \ldots, \hat{p}(\hat{m}_c)) := \mathbf{p}^*(\hat{\theta}_c, \hat{m}_c).$$

The gain-seeking-customer pricing policy $\mathcal{G}$ operates in episodes of stochastically increasing length indexed by $c \in \mathbb{N} \cup \{0\}$. Episode c starts in period $\tau_{c-1} + 1$ and ends in period τ_c. At the beginning of episode c, the most recent estimate of the optimal price cycle, $\hat{p} = (\hat{p}(1), \ldots, \hat{p}(\hat{m}_c))$, is repeated c times. After that, the policy checks if the smallest eigenvalue of the Fisher information matrix $\mathcal{J}^{\mathcal{G}}_{\tau_{c-1}}$ exceeds $\sqrt{\tau_{c-1}}$. If this is not the case, then four test prices (namely, $\tilde{p}_1, \tilde{p}_2, \tilde{p}_3$, and $\tilde{p}_4$) are charged to ensure the consistency of demand parameter estimates. At the end of the episode, the estimates of the demand parameters and the optimal price cycle are updated using the recently acquired sales data, and the policy proceeds with the next episode.

The next result provides two upper bounds on the T-period regret of the policy $\mathcal{G}$.

Theorem 6 (Asymptotic Optimality of the Gain-Seeking Customer Policy). *Let $\rho \in \mathcal{R}_0$, $\theta \in int(\Theta)$, and $\pi = \mathcal{G}$. Suppose that $m^*(\cdot)$ is continuous at θ. Then, there exists a finite and positive constant $\check{c}_2$ satisfying the following: if $m^*(\theta) \geq 3$, then*

$$\Delta_\pi(T, \rho, \theta) \leq \check{c}_2 \log^2 T \quad \textit{for all} \quad T \in \mathbb{N}, \tag{5.8}$$

whereas if $m^*(\theta) = 2$, then

$$\Delta_\pi(T, \rho, \theta) \leq \check{c}_2 \sqrt{T} \log T \quad \textit{for all} \quad T \in \mathbb{N}. \tag{5.9}$$

Theorem 6 provides theoretical performance guarantees for the policy $\mathcal{G}$. If the length of the optimal price cycle exceeds two, then this policy has logarithmically growing regret, whereas if the length of the optimal price cycle equals two, then the regret of the same policy grows in the order of $\sqrt{T}$ (up to logarithmic terms). The following result shows that these performance guarantees are asymptotically optimal.

Theorem 7 (Lower Bound on Regret for Gain-Seeking Customers) *Let $\rho \in \mathcal{R}_0$. Assume that $\epsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, where $\sigma > 0$. Then, there exists a finite and positive constant $\check{c}_3$ such that, under any admissible policy π,*

$$\sup_{\theta \in \Theta,\, m^*(\theta) \geq 3} \{\Delta_\pi(T, \rho, \theta)\} \geq \check{c}_3 \log T \quad \textit{for all } T \in \mathbb{N}, \tag{5.10}$$

and

$$\sup_{\theta \in \Theta,\, m^*(\theta) = 2} \{\Delta_\pi(T, \rho, \theta)\} \geq \check{c}_3 \sqrt{T} \quad \textit{for all } T \in \mathbb{N}. \tag{5.11}$$

The preceding two theorems show that, in the case of gain-seeking customers, the best achievable regret performance depends on the length of the optimal price cycle. In particular, if the length of the optimal price cycle exceeds two, the performance improves in the sense that the seller can achieve logarithmically growing regret. This is because the natural price variation within the optimal price cycle induces a rapid learning rate in this case. On the other hand, if the length of the optimal price cycle is two, there is simply not enough price variation to induce the same learning rate.

We note that the asymptotically optimal performance guarantees in Theorem 6 are valid for all θ at which $m^*(\cdot)$ is continuous. Numerical evidence strongly suggests that this only excludes a set of (Lebesgue) measure zero, and therefore, only little generality is lost: for example, if one assumes that θ is drawn from Θ according to an absolutely continuous probability distribution, then with probability one, θ is a continuity point of $m^*(\cdot)$ so that Theorem 6 is valid. (In 100,000 randomly sampled parameters $\theta = (\alpha, \beta, \eta)$ with (α, β) drawn uniformly from $[0,1]$ and η drawn uniformly from $[0, 2\beta]$, we found no single instance with $R^*_{m^*(\theta)} = R^*_m(\theta)$ for some $m \neq m^*(\theta)$, implying that, in all 100,000 instances, $m^*(\cdot)$ is continuous at θ.)

6. Statistical Test for Loss Aversion

There is strong evidence that customers often behave in a loss-averse manner—see, for example, the empirical research literature referred to in Section 1.3. However, it may not always be clear up front whether loss aversion is present in a specific market. In this section, we design and analyze a statistical test for detecting loss aversion based solely on sales transaction data.

We consider the base setting in Section 2. The test statistic for detecting loss aversion uses the transaction data gathered at the beginning of the selling horizon in the following manner.

Test Statistic for Detecting Loss Aversion

Let $n \in \mathbb{N}$. For $t = 1, \ldots, 4n$, set prices as follows:

For $t = 1, \ldots, n$, choose $p_t = \underline{p}$.

For $t = n+1, \ldots, 2n$, choose $p_t = \overline{p}$.

For $t = 2n+1, \ldots, 4n$, choose $p_t = \underline{p}$ if t is odd and $p_t = \overline{p}$ if t is even.

Compute

$X_{1,n} = \frac{1}{n}\sum_{t=1}^{n} d_t$, $X_{2,n} = \frac{1}{n}\sum_{t=n+1}^{2n} d_t$, $X_{3,n} = \frac{1}{2n}\sum_{t=2n+1}^{4n} d_t$, and

$$S_n = X_{3,n} - X_{1,n} - X_{2,n}.$$

Based on the data gathered in the initial $4n$ periods, we propose the following simple test: use the slow-moving policy only if $S_n < 0$. The next proposition shows that, when customers are loss-averse, the preceding test detects loss aversion with high probability.

Proposition 6 (False Negative Error) *If $\eta^+ < \eta^-$, then there exists a constant $\overline{c}_1$ such that*

$$\mathbb{P}\{S_n \geq 0\} \leq \overline{c}_1/n \text{ for all } n \in \mathbb{N}.$$

In the proof of Proposition 6, we show that the expected value of S_n is proportional to $\eta^+ - \eta^-$ and also has a term that depends on the (unknown) initial reference price r_1 and that vanishes at a rate of order $\frac{1}{n}$. This explicit expression of the expected value of S_n and especially its proportionality to $\eta^+ - \eta^-$ follows from the careful design of the prices $\{p_1, \ldots, p_{4n}\}$. Using a suitable concentration inequality, we deduce that the probability that the seller fails to detect loss-averse behavior is of order $\frac{1}{n}$.

Choosing $n = \lceil \sqrt{T} \rceil$, it follows from Proposition 6 that the probability of failing to detect loss-averse behavior when it is present is at most in the order of $1/\sqrt{T}$. In this case, the corresponding regret in the subsequent periods clearly is at most linear in T (regardless of the seller's policy). On the other hand, if the test successfully detects loss aversion and the slow-moving policy from Section 3 is used in the subsequent periods, Theorem 2 guarantees a regret of order $\sqrt{T}$. Combining both cases, it follows that applying the statistical test prior to using the slow-moving pricing policy keeps the regret at the asymptotically optimal rate of $\sqrt{T}$.

For the sake of completeness, we also provide a result on the probability of falsely detecting loss-averse behavior.

Proposition 7 (False Positive Error). *If $\eta^+ > \eta^-$, then there is a constant $\overline{c}_2$ such that*

$$\mathbb{P}\{S_n < 0\} \leq \overline{c}_2/n \text{ for all } n \in \mathbb{N}.$$

The proof of this result is almost entirely identical to that of Proposition 6 and is therefore omitted.

7. Concluding Remarks

Motivated by the strong evidence in behavioral economics and marketing literatures that reference effects play a crucial role in consumers' purchasing decisions, we study in this paper the learn-and-earn trade-off in dynamic pricing with an unknown demand function and reference effects. In the case of loss-averse customers, we design and analyze a pricing policy that changes prices very slowly to control the evolution of the reference effects while implementing price experiments with carefully tuned frequency and magnitude to optimally balance learning and earning. We prove that our policy has asymptotically optimal revenue performance even though it does not attempt to learn the evolution of the reference-price process. We also extend these results to more general time-varying reference prices. Furthermore, in a case in which the reference price is fixed, we design and analyze another policy that uses a judicious amount of price experimentation, thereby achieving asymptotically optimal performance. In this case, we also prove a strong incomplete-learning result that shows that a policy without forced price experiments may incur large losses. Finally, we extend our analysis from the case of loss-averse customers to the case of gain-seeking customers, and we design statistical tests that the seller can use to validate different modeling assumptions. We hope that our analysis stimulates further study on the impact of behavioral aspects on dynamic estimation-and-pricing policies.

Our results in Section 3 are concerned with reference-price updating mechanisms based on exponential smoothing and its generalizations (as in Section 3.3). This reference-price updating model has been used extensively in the existing literature in both empirical (see, e.g., Lattin and Bucklin 1989, Hardie et al. 1993, Kalyanaram and Little 1994) and theoretical (see, e.g., Greenleaf 1995, Kopalle et al. 1996, Fibich et al. 2003, Popescu and Wu 2007) studies. As a possible avenue for future research, we suggest the study of dynamic pricing and demand learning with alternative reference-price updating mechanisms, such as reference-price processes with longer memory.

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